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See CNAD AC/225 STANAG distribution

**STANAG 4355 JAIS (EDITION 3) – THE MODIFIED POINT MASS AND FIVE DEGREES OF FREEDOM TRAJECTORY MODELS**

References:

- a. MAS/24-LAND/4355 dated 20 January 1997 (Edition 2)
- A. PFP(NAAG)D(2006)0005, dated 21 March 2006, (Edition 3) (Ratification Draft)

1. The enclosed NATO Standardization Agreement, which has been ratified by nations as reflected in the NATO Standardization Document Database (NSDD) is promulgated herewith.
2. The references listed above are to be destroyed in accordance with local document destruction procedures.

ACTION BY NATIONAL STAFFS

3. National staffs are requested to examine their ratification status of the STANAG and, if they have not already done so, advise the Defence Investment Division through their national delegation as appropriate of their intention regarding its ratification and implementation.

A handwritten signature in black ink, appearing to read 'Juan A. Moreno', is written over a horizontal line.

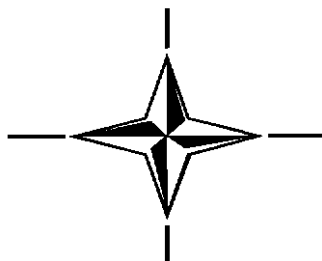
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Enclosure:

STANAG 4355 (Edition 3)

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**NORTH ATLANTIC TREATY ORGANISATION  
(NATO)**

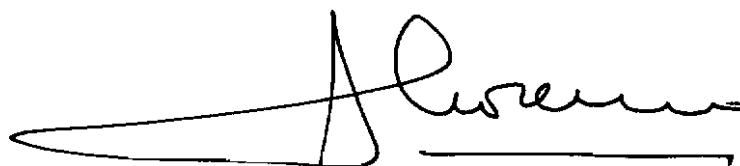


**NATO STANDARDIZATION AGENCY  
(NSA)**

**STANDARDISATION AGREEMENT  
(STANAG)**

SUBJECT: THE MODIFIED POINT MASS AND FIVE DEGREES OF FREEDOM  
TRAJECTORY MODELS

Promulgated on 17 April 2009



Juan A. MORENO  
Vice-Admiral, ESP(N)  
Director, NATO Standardization Agency

RECORD OF AMENDMENTS

| No. | Reference/date of Amendment | Date Entered | Signature |
|-----|-----------------------------|--------------|-----------|
|     |                             |              |           |

EXPLANATORY NOTES

AGREEMENT

1. This STANAG is promulgated by the Director NATO Standardization Agency under the authority vested in him by the NATO Standardization Organisation Charter.
2. No departure may be made from the agreement without informing the tasking authority in the form of a reservation. Nations may propose changes at any time to the tasking authority where they will be processed in the same manner as the original agreement.
3. Ratifying nations have agreed that national orders, manuals and instructions implementing this STANAG will include a reference to the STANAG number for purposes of identification.

RATIFICATION, IMPLEMENTATION AND RESERVATIONS

4. Ratification, implementation and reservation details are available on request or through the NSA websites (internet <http://nsa.nato.int>; NATO Secure WAN <http://nsa.hq.nato.int>).

FEEDBACK

5. Any comments concerning this publication should be directed to NATO/NSA Bvd Leopold III- 1110 Brussels - Belgium.

NATO STANDARDISATION AGREEMENT  
(STANAG)

THE MODIFIED POINT MASS AND FIVE DEGREES OF FREEDOM TRAJECTORY MODELS

ANNEXES:

- A Forms of Fire Control Input Data
- B Additional Terms for Projectiles with Bourrelet Nubs
- C Additional Terms for Spin-Stabilized Rocket-Assisted Projectiles, Spin-Stabilized Base-Burn Projectiles and Fin-Stabilized Rockets – Method 1 –
- D Additional Terms for Spin-Stabilized Rocket-Assisted Projectiles, Spin-Stabilized Base-Burn Projectiles and Fin-Stabilized Rocket-Assisted Rockets – Method 2
- E Calculation of Submunition Trajectories
- F Point Mass Trajectory Model
- G Additional Terms for Guided Munitions
- H Glossary of Terms
- I Selected Bibliography

Related Documents:

- AOP-37 - NATO Armaments Ballistic Kernel (NABK)
- ISO 2533-1975 The ISO Standard Atmosphere
- STANAG 4106 - Procedures to Determine the Degree of Ballistic Performance Similarity of NATO Indirect Fire Ammunition and the Applicable Corrections to Aiming Data
- STANAG 4114 - Procedures to Determine the Fire Control Inputs for Use in Indirect Fire Control Systems
- STANAG 4119 - Adoption of a Standard Cannon Artillery Firing Table Format
- STANAG 4537 NATO Armaments Ballistic Kernel (NABK)
- STANAG 6022 - Adoption of a Standard Gridded Data Meteorological Message

## AIM

1. The principal aim of this agreement is to standardize the exterior ballistic trajectory simulation methodology for NATO Naval and Army Forces. The Modified Point Mass model will be used for spin-stabilized projectiles and the Five Degrees of Freedom model will be used for fin-stabilized rockets. This will facilitate the exchange of exterior ballistic data and fire control information.

## AGREEMENT

2. Participating nations agree to use the Modified Point Mass Trajectory Model for spin-stabilized projectiles and a Five Degrees of Freedom Model for exterior ballistic trajectory simulation of fin-stabilized rockets.

## GENERAL

3. This agreement permits some flexibility by accommodating certain specific national aerodynamic conventions and ballistic data analysis procedures.

## DETAILS OF AGREEMENT

4. The details of the agreement are given hereunder and are divided into the following four parts:

- I - Equations of Motion
  - A. Equations of Motion For Spin-Stabilized Projectiles
  - B. Equations of Motion for Fin-Stabilized Rockets
  - C. Common Equations of Motion
- II - List of Symbols
- III - Comparison of Aerodynamic Coefficient Symbols
- IV - List of Data Requirements

## IMPLEMENTATION OF AGREEMENT

5. This STANAG is considered to be implemented when a nation has issued the necessary orders/instructions to the forces concerned, putting the procedures detailed in this agreement into effect.

## I. EQUATIONS OF MOTION

5. The following equations constitute mathematical models representing the flight of:

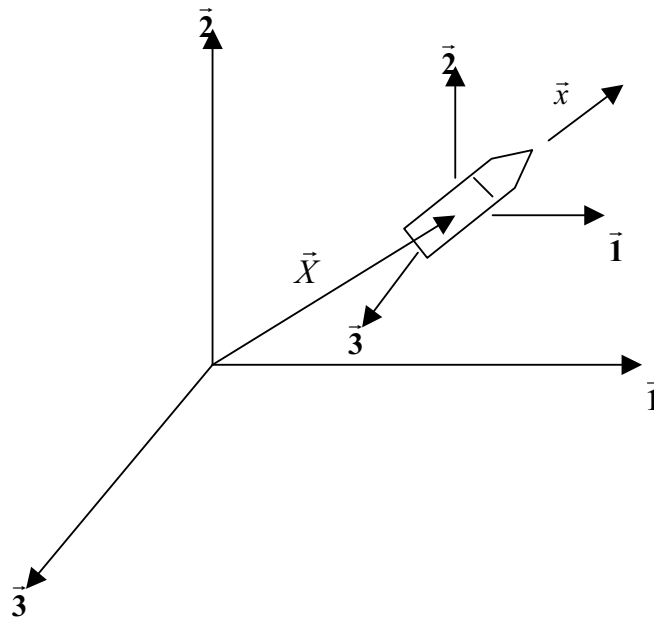
- (1) spin-stabilized projectiles and
- (2) fin-stabilized rockets,

that are dynamically stable and possess at least trigonal symmetry. The mathematical modeling is accomplished mainly by: (a) including only the most essential forces and moments, (b) for spin-stabilized projectiles, approximating the actual yaw by the yaw of repose neglecting transient yawing motion, and (c) applying fitting factors to some of the above forces to compensate for the neglect of or approximations for other forces and moments. All vectors have as a frame of reference a right-handed, orthonormal, ground-fixed, Cartesian coordinate system with unit vectors ( $\vec{1}$ ,  $\vec{2}$  and  $\vec{3}$ ) as shown in Figure I-1.

Assume that the body can be considered a solid of revolution, and assign to the axis of rotational symmetry a unit vector  $\vec{x}$  in the chosen coordinate system. Since the rigid body is to represent a projectile or rocket, the direction of  $\vec{x}$  from tail to nose is defined as positive. The total angular momentum of the body can now be expressed as the sum of two vectors in the ground-fixed coordinate system:

- (1) The angular momentum about  $\vec{x}$ , and
- (2) the angular momentum about an axis perpendicular to  $\vec{x}$ .

The angular momentum about  $\vec{x}$  has the magnitude ( $I_x p$ ); where  $I_x$  is the moment of inertia of the body about  $\vec{x}$ , and  $p$  is the axial spin or angular velocity about  $\vec{x}$ , in radians per second. Therefore, the total angular momentum about  $\vec{x}$  can be represented by the vector ( $I_x p \vec{x}$ ). In this document, a positive  $p$  is defined as rotation, which would cause a right-hand screw to advance in the direction of  $\vec{x}$ .



**Figure I-1.** Cartesian Coordinate System with Unit Vectors

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Edition 3

The total angular velocity of the body about an axis perpendicular to  $\vec{x}$  is given by the vector  $(\vec{x} \times \dot{\vec{x}})$ , where the superscript dot refers to differentiation with respect to time. Since the body possesses rotational symmetry, every axis through the center-of-mass and perpendicular to  $\vec{x}$  is a principle axis of inertia. If the moment of inertia of the body about any transverse axis is  $I_Y$ , the total angular momentum about an axis perpendicular to  $\vec{x}$  has the vector representation  $I_Y (\vec{x} \times \dot{\vec{x}})$ .

Let  $\vec{H}$  denote the total vector angular momentum of the body. The vector representation of  $\vec{H}$  is:

$$\vec{H} = I_X p \vec{x} + I_Y (\vec{x} \times \dot{\vec{x}}) \quad (a)$$

Let  $\sum \vec{M}$  be the sum of the vector applied moments, and set  $\sum \vec{M}$  equal to the vector rate of change of angular momentum.

$$\dot{\vec{H}} = \sum \vec{M} \quad (b)$$

In addition to these basic equations, two additional expressions are needed for use in the force-moment system.

$$(\vec{H} \times \vec{x}) = I_Y \dot{\vec{x}} \quad (c)$$

$$(\vec{H} \cdot \vec{x}) = I_X p \quad (d)$$

Equation (b) is the basic vector differential equations of angular motion in a fixed coordinate system. The basic equation of motion for the center-of-mass is:

$$\dot{\vec{u}} = \frac{\sum \vec{F}}{m} \quad (e)$$

where  $\sum \vec{F}$  denotes the sum of the vector applied forces,  $m$  is the mass of the body, and  $\dot{\vec{u}}$  is the vector acceleration of the center-of-mass in the fixed coordinate system.

A. EQUATIONS OF MOTION FOR SPIN-STABILIZED PROJECTILES

6. Newton's law of motion of the center of mass of the projectile is:

$$\vec{F} = m \dot{\vec{u}} = \overrightarrow{DF} + \overrightarrow{LF} + \overrightarrow{MF} + m \vec{g} + m \vec{A} \quad (1)$$

where acceleration due to drag force, when  $i$  is used as a fitting factor, is:

$$\frac{\overrightarrow{DF}}{m} = - \left( \frac{\pi \rho i d^2}{8m} \right) \left( C_{D_0} + C_{D_{\alpha^2}} (Q_D \alpha_e)^2 + C_{D_{\alpha^4}} (Q_D \alpha_e)^4 \right) v \vec{v} \quad (2)$$

or, when  $C$  is used as a fitting factor,

$$\frac{\overrightarrow{DF}}{m} = - \left( \frac{\pi \rho m_r}{8Cm} \right) \left( C_{D_0} + C_{D_{\alpha^2}} (Q_D \alpha_e)^2 + C_{D_{\alpha^4}} (Q_D \alpha_e)^4 \right) v \vec{v} \quad (3)$$

or, when  $f_D$  is used as a fitting factor,

$$\frac{\overrightarrow{DF}}{m} = - \left( \frac{\pi \rho d^2}{8m} \right) \left( f_D C_{D_0} + C_{D_{\alpha^2}} (Q_D \alpha_e)^2 + C_{D_{\alpha^4}} (Q_D \alpha_e)^4 \right) v \vec{v} \quad (4)$$

acceleration due to lift force is:

$$\frac{\overrightarrow{LF}}{m} = \left( \frac{\pi \rho d^2 f_L}{8m} \right) \left( C_{L_{\alpha}} + C_{L_{\alpha^3}} \alpha_e^2 + C_{L_{\alpha^5}} \alpha_e^4 \right) v^2 \vec{\alpha}_e \quad (5)$$

acceleration due to Magnus force is:

$$\frac{\overrightarrow{MF}}{m} = - \frac{\pi \rho d^3 Q_M p C_{mag-f}}{8m} (\vec{\alpha}_e \times \vec{v}) \quad (6)$$

acceleration due to gravity ( $g$ ) is:

(See Section I-C, Common Equations of Motion)

acceleration due to the Coriolis effect ( $\Lambda$ ) is:

(See Section I-C, Common Equations of Motion)



7. The magnitude of spin acceleration is given by:

$$\dot{p} = \frac{\pi \rho d^4 p v C_{spin}}{8 I_x} \quad (7)$$

where:

$$p = p_0 + \int_0^t \dot{p} dt \quad (8)$$

is the magnitude of spin at time = t, and

$$p_0 = \frac{2\pi u_0}{t_c d} \quad (9)$$

is the magnitude of the initial spin of the projectile at the muzzle.

8. The yaw of repose is given by:

$$\vec{\alpha}_e = - \frac{8 I_x p (\vec{v} \times \dot{\vec{u}})}{\pi \rho d^3 (C_{M_\alpha} + C_{M_{\alpha^3}} \alpha_e^2) v^4} \quad (10)$$

where:

$$\vec{\alpha}_{e_0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (11)$$

is the initial value of the yaw of repose.

To begin the numerical integration of the modified point mass equations of motion using a predictor-corrector method, one must predict the value of  $\vec{u}$  using  $\vec{\alpha}_{e_0}$  (Eq. (11)) and then use this value of  $\vec{u}$  to calculate  $\vec{\alpha}_e$  using Eq. (10).

9. See section I-C for the velocity of the projectile with respect to the ground fixed axis at time = t.  
See section I-C for the velocity of the projectile with respect to the air at time = t.  
See section I-C for the position of the projectile with respect to the ground axis at time = t.

10. Windage jump, the correction for wind shear between successive integration steps, is given as a velocity correction:

$$\Delta \vec{u} = \frac{\left( C_{L_\alpha} + C_{L_{\alpha^3}} \alpha_e^2 + C_{L_{\alpha^5}} \alpha_e^4 \right) f_L I_X p (\vec{u} \times \Delta \vec{w})}{(C_{M_\alpha} + C_{M_{\alpha^3}} \alpha_e^2) m d v^2} \quad (12)$$

where:  $\Delta \vec{w} = \vec{w}_t - \vec{w}_{t-\Delta t}$  (13)  
and at  $t = 0$ ,  $\vec{w}_t = \vec{w}_0 = 0$ .

11. See section I-C for the position of the projectile with respect to the spherical earth's surface.

## B. EQUATIONS OF MOTION FOR FIN-STABILIZED ROCKETS

The equations of motion for fin-stabilized rockets are described during four phases of the flight, namely, during launch, from launch to fin opening, from fin opening to the end of motion requiring five degrees of freedom (5 DoF), and from the end of the 5 DoF segment to the end of the 3 DoF phase which normally is the burst point of the warhead. Motor will normally be functioning from  $t_0$  until before (near) the end of Phase 3, Time of End of Five Degrees of Freedom ( $t_0 \leq t < t_{E5D}$ ). These phases were chosen to represent those portions of the motion, which require significantly different force and moment descriptions to properly simulate actual flight conditions.

### **Phase 1: Time Zero to Time of Launch** ( $t_0 \leq t \leq t_L$ )

#### Initial Launching Phase

12. The initial position of the center-of-mass,  $\vec{X}_0$ , with-respect-to the ground-fixed axis, acceleration, and angular momentum for a rocket are as follows:

The drag force ( $\overrightarrow{DF}$ ), thrust force ( $\overrightarrow{TF}$ ), acceleration due to gravity and the deceleration due to rocket-launcher friction are considered during the rocket-launching phase. The equation of motion of the center-of-mass during the rocket launch phase is given by:

$$\vec{F} = m \dot{\vec{u}} = (\overrightarrow{DF} + \overrightarrow{TF}) + m|\vec{g}|(-\sin QE - \mu_f \cos QE)\vec{x}_0 \quad (14)$$

NOTE: In this phase the rocket does not move before  $t = t_{FM}$  and is restricted to forward motion, i.e., for Eq. (14):

$$\vec{F} \equiv \vec{0} \text{ for } t \leq t_{FM} \text{ or whenever } \vec{F} < \vec{0}$$

where acceleration due to drag force is:

$$\frac{\overrightarrow{DF}}{m} = -\left(\frac{\pi i \rho d^2}{8 m}\right) C_{D_0} v^2 \vec{x}_0 \quad (15)$$

acceleration due to thrust force is:

$$\frac{\overrightarrow{TF}}{m} = \frac{(f_T \dot{m}_f I_{SP} + (P_r - P) A_e)\vec{x}_0}{m} \quad (16)$$

where:

$$\vec{x}_0 = \begin{bmatrix} \cos QE \cos \Delta AZ \\ \sin QE \\ \cos QE \sin \Delta AZ \end{bmatrix} \quad (17)$$

and acceleration due to gravity ( $\vec{g}$ ) and  $\vec{X}_0$  are given in Section I-C.

Initial Angular Momentum at Time of Launch ( $t_L$ )

13. At  $t_L$ , the initial angular momentum of the rocket is assigned.  $p_{(t_L)}$  is the initial angular velocity (spin) of the rocket about  $\vec{x}$ , and  $\omega_{y(t_L)}$  and  $\omega_{z(t_L)}$  are the transverse angular velocities of the rocket at  $t_L$ .  $\omega_{y(t_L)}$  and  $\omega_{z(t_L)}$  are used to match the observed angular performance of the rocket. The initial value of  $\vec{H}_{(t_L)}$  is given by:

$$\vec{H}_{(t_L)} = I_{X(t_L)} p_{(t_L)} \vec{x}_{(t_L)} + I_{Y(t_L)} \begin{bmatrix} -\omega_{y(t_L)} \sin QE \cos \Delta AZ - \omega_{z(t_L)} \sin \Delta AZ \\ + \omega_{y(t_L)} \cos QE \\ -\omega_{y(t_L)} \sin QE \sin \Delta AZ + \omega_{z(t_L)} \cos \Delta AZ \end{bmatrix} \quad (18)$$

where:

$$\vec{x}_{(t_L)} = \vec{x}_0 \quad (19)$$

**Phase 2: Time of Launch to Time of Rocket Fins Opening ( $t_L \leq t \leq t_{FO}$ )**

Use aerodynamics for rockets with fins closed.

Equation of Motion of the Center-of-Mass of the Body

14. The motion of the center-of-mass of the body is a summation of the accelerations acting on the body and is given by:

$$\vec{F} = m \dot{\vec{u}} = \overrightarrow{DF} + \overrightarrow{LF} + \overrightarrow{MF} + \overrightarrow{PDF} + \overrightarrow{TF} + m \vec{g} + m \vec{\Lambda} \quad (20)$$

where:

acceleration due to drag forces is:

$$\frac{\overrightarrow{DF}}{m} = - \left( \frac{\pi \rho d^2}{8m} \right) \left( C_{D_0} + C_{D_{\alpha^2}} (\alpha)^2 \right) v \vec{v} \quad (21)$$

acceleration due to lift forces is:

$$\frac{\overrightarrow{LF}}{m} = \left( \frac{\pi \rho d^2 f_L}{8m} \right) \left( C_{L_\alpha} + C_{L_{\alpha^3}} (\alpha)^2 \right) \left( v^2 \vec{x} - (\vec{v} \cdot \vec{x}) \vec{v} \right) \quad (22)$$

acceleration due to Magnus forces is:

$$\frac{\overrightarrow{MF}}{m} = - \frac{\pi \rho d^3 C_{mag-f}}{I_X 8m} (\vec{H} \cdot \vec{x}) (\vec{x} \times \vec{v}) \quad (23)$$

acceleration due to pitch damping force is:

$$\frac{\overrightarrow{PDF}}{m} = \left( \frac{\pi \rho d^3 (C_{N_q} + C_{N_{\dot{\alpha}}})}{I_Y 8 m} \right) v (\vec{H} \times \vec{x}) \quad (24)$$

acceleration due to thrust force is:

$$\frac{\overrightarrow{TF}}{m} = \frac{(f_T \dot{m}_f I_{SP} + (P_r - P) A_e) \vec{x}}{m} \quad (25)$$

acceleration due to gravity (  $g$  ) is:

(See Section I-C, Common Equations of Motion)

acceleration due to the Coriolis effect (  $\Lambda$  ) is:

(See Section I-C, Common Equations of Motion)

NOTE: (See Annex C or D for  $\dot{m}_f$  )

#### Equation of Angular Momentum of the Body

15. The angular momentum of the body is the summation of the moments acting on the body and is given by:

$$\dot{\vec{H}} = \overrightarrow{OM} + \overrightarrow{PDM} + \overrightarrow{MM} + \overrightarrow{SDM} + \overrightarrow{FCM} + \overrightarrow{AJDM} + \overrightarrow{TJDM} \quad (26)$$

where:

Angular momentum due to Overturning Moment:

$$\overrightarrow{OM} = \left( \frac{\rho d^3 \pi}{8} \right) (C_{M_\alpha} + C_{M_{\alpha^3}} \alpha^2) v (\vec{v} \times \vec{x}) \quad (27)$$

Angular momentum due to Pitch Damping Moment:

$$\overrightarrow{PDM} = \left( \frac{\rho d^4 \pi}{8 I_Y} \right) (C_{M_q} + C_{M_{\dot{\alpha}}}) v [\vec{H} - (\vec{H} \cdot \vec{x}) \vec{x}] \quad (28)$$

Angular momentum due to Magnus Moment:

$$\overrightarrow{MM} = - \left( \frac{\rho d^4 \pi}{8 I_X} \right) C_{mag-m} (\vec{H} \cdot \vec{x}) [(\vec{v} \cdot \vec{x}) \vec{x} - \vec{v}] \quad (29)$$

Angular momentum due to Spin Damping Moment:

$$\overrightarrow{SDM} = \left( \frac{\rho d^4 \pi}{I_X 8} \right) C_{spin} v (\vec{H} \cdot \vec{x}) \vec{x} \quad (30)$$

Angular momentum due to Fin Cant Moment:

$$\overrightarrow{FCM} = \left( \frac{\rho d^3 \pi}{8} \right) C_{l_\delta} \varepsilon v^2 \vec{x} \quad (31)$$

Angular momentum due to Axial Jet Damping Moment:

$$\overrightarrow{AJDM} = \left( \frac{\dot{m} r_{ne}^2}{2 I_X} \right) (\vec{H} \cdot \vec{x}) \vec{x} \quad (32)$$

Angular momentum due to Transverse Jet Damping Moment:

$$\overrightarrow{TJDM} = \left( \frac{\dot{m} r_e r_t}{I_Y} \right) [\vec{H} - (\vec{H} \cdot \vec{x}) \vec{x}] \quad (33)$$

NOTE: (See Annex C or D for effects of changes in mass during burning.)

The angular momentum and unit vector along the longitudinal axis at time =  $t$  is:

$$\vec{H} = \vec{H}_{(t_L)} + \int_{t_L}^t \dot{\vec{H}} dt \quad (34)$$

$$\vec{x} = \vec{x}_{(t_L)} + \int_{t_L}^t \dot{\vec{x}} dt \quad (35)$$

where  $\vec{H}_{(t_L)}$  is defined at Eq. (18), and

$$\vec{x}_{(t_L)} = \vec{x}_0 \quad (36)$$

and

$$\dot{\vec{x}} = \frac{(\vec{H} \times \vec{x})}{I_Y} \quad (37)$$

NOTE: While integrating the equations of motion during the 5 DoF phase, the magnitude of the unit vector  $\vec{x}$  should be normalized to the value 1.

The yaw of the symmetric rigid-body ( $\alpha$ ) is the included angle between  $\vec{x}$  and  $\vec{v}$ . The yaw is defined as a positive quantity and the magnitude is given by:

$$\alpha = \arccos \left[ \frac{(\vec{v} \cdot \vec{x})}{|\vec{v}|} \right] \quad (38)$$

The orientation of the yaw ( $\psi$ ) is the angle of the plane containing  $\vec{v}$  and  $\vec{x}$  relative to the vertical plane containing  $\vec{v}$  measured in a clockwise direction. The orientation of the yaw is given by:

$$\psi = \arctan \left[ \frac{(v_1 x_3 - v_3 x_1) v}{v_1 (v_1 x_2 - v_2 x_1) - v_3 (v_2 x_3 - v_3 x_2)} \right] \quad (39)$$

**Phase 3: Time from Rocket Fins Opening to Time of End of Five Degrees of Freedom ( $t_{FO} \leq t \leq t_{ESD}$ )**

16. Same as equations in ( $t_L \leq t \leq t_{FO}$ ) with aerodynamics for rocket with fins open. Define  $t_{ESD}$  to begin after a small time interval after motor burnout ( $t_{BO}$ ), to allow for aerodynamic settling.

**Phase 4: Time from End of Five Degrees of Freedom ( $t \leq t_{ESD}$ )**

17. Use point mass equations.

The center of mass of the projectile is:

$$\vec{F} = m \dot{\vec{u}} = \overrightarrow{DF} + m \vec{g} + m \vec{\Lambda} \quad (40)$$

where acceleration due to drag forces is:

$$\frac{\overrightarrow{DF}}{m} = - \left( \frac{\pi \rho i d^2}{8m} \right) C_D v \vec{v} \quad (41)$$

NOTE: ( $\vec{g}$ ,  $\vec{u}$ , and  $\vec{\Lambda}$  are defined in section I-C)

C. COMMON EQUATIONS OF MOTION

18. Acceleration due to gravity is:

$$\vec{g} = -g_0 \left( R^2 / r^3 \right) \vec{r} = -g_0 \begin{bmatrix} \frac{X_1}{R} \\ \frac{X_2}{R} \\ \frac{X_3}{R} \end{bmatrix} \quad (42)$$

$$\text{where: } g_0 = 9.80665 [1 - .0026 \cos(2 \text{ lat})] \quad (43)$$

$$\vec{r} = \vec{X} - \vec{R}, \quad (44)$$

$$\vec{R} = \begin{bmatrix} 0 \\ -R \\ 0 \end{bmatrix} \quad (45)$$

$$\text{and } R = 6.356766 \times 10^6 \text{ meters}$$

19. Acceleration due to the effect of the Coriolis force is:

$$\vec{\Lambda} = -2(\vec{\omega} \times \vec{u}) \quad (46)$$

where:

$$\vec{\omega} = \begin{bmatrix} \Omega \cos(\text{lat}) \cos(AZ) \\ \Omega \sin(\text{lat}) \\ -\Omega \cos(\text{lat}) \sin(AZ) \end{bmatrix}, \quad (47)$$

$$\text{and } \Omega = 7.292115 \times 10^{-5} \text{ rad/s}$$

NOTE: For Southern Hemisphere, *lat* is negative.

20. The velocity of the projectile or rocket with respect to the ground fixed axes at time = *t* is:

$$\vec{u} = \vec{u}_0 + \int_0^t \dot{\vec{u}} dt \quad (48)$$

where:

$$\vec{u}_0 = \begin{bmatrix} u_0 \cos(QE) \cos(\Delta AZ) \\ u_0 \sin(QE) \\ u_0 \cos(QE) \sin(\Delta AZ) \end{bmatrix} \quad (49)$$



21. The velocity of the projectile or rocket with respect to the air is given by:

$$\vec{v} = \vec{u} - \vec{w} \quad (50)$$

and Mach number:

$$M = v/a \quad (51)$$

where the speed of sound,  $a$ , is:

$$a = \left[ \frac{\gamma P}{\rho} \right]^{1/2} = 20.046796 (T_v + 273.15)^{1/2} \quad (52)$$

$\gamma$  = ratio of specific heats for air

$P$  = air pressure

$$\rho = \text{air density} = \frac{0.003483678761P}{T_v + 273.15} \quad (53)$$

$T_v$  = virtual temperature

NOTE: Values used in (52) and (53) referenced from STANAG 6022

22. The position of the projectile or rocket with respect to the ground axes is given by:

$$\vec{X} = \vec{X}_0 + \int_0^t \vec{u} dt \quad (54)$$

where:

$$\vec{X}_0 = \begin{bmatrix} l_w \cos(QE) \cos(\Delta AZ) \\ X_{2w} + l_w \sin(QE) \\ l_w \cos(QE) \sin(\Delta AZ) \end{bmatrix} \quad (55)$$

is the location of the muzzle, i.e., initial position of the center of mass of the projectile or rocket. For a rocket,  $l_w = 0$ .

23. The position of the projectile or rocket with respect to the spherical earth's surface is given by the approximation:

$$\vec{E} = \begin{bmatrix} X_1 \\ X_2 + \frac{(X_1^2 + X_3^2)}{2R} \\ X_3 \end{bmatrix} \quad (56)$$

II. LIST OF SYMBOLS

| <u>Symbol</u>           | <u>Definition</u>                                                      | <u>Units</u>       |
|-------------------------|------------------------------------------------------------------------|--------------------|
| $a$                     | Local speed of sound in air                                            | $m/s$              |
| $A_e$                   | Exit area of the motor jet                                             | $m^2$              |
| $\overrightarrow{AJDM}$ | Angular momentum due to axial jet damping moment                       | $kg\ m^2\ rad/s^2$ |
| $AZ$                    | Azimuth (Bearing) of $\vec{1}$ axis measured clockwise from true North | $mil\ ^*$          |
| $C$                     | Ballistic coefficient                                                  | $kg/m^2$           |
| $C_D$                   | Drag force coefficient                                                 | $none$             |
| $C_{D_0}$               | Zero-yaw drag force coefficient                                        | $none$             |
| $C_{D_{\alpha^2}}$      | Quadratic drag force coefficient                                       | $1/rad^2\ ^{**}$   |
| $C_{D_{\alpha^4}}$      | Quartic drag force coefficient                                         | $1/rad^4$          |
| $C_{L_\alpha}$          | Lift force coefficient                                                 | $1/rad$            |
| $C_{L_{\alpha^3}}$      | Cubic lift force coefficient                                           | $1/rad^3$          |
| $C_{L_{\alpha^5}}$      | Quintic lift force coefficient                                         | $1/rad^5$          |
| $C_{l_\delta}$          | Fin cant moment coefficient                                            | $1/rad$            |
| $C_{M_\alpha}$          | Overturning moment coefficient                                         | $none$             |
| $C_{M_{\alpha^3}}$      | Cubic overturning moment coefficient                                   | $1/rad^2$          |
| $C_{mag-f}$             | Magnus force coefficient                                               | $1/rad^2$          |
| $C_{mag-m}$             | Magnus moment coefficient                                              | $1/rad$            |

---

\* 1 mil = 1/6400 of a circle =  $2\pi/6400$  radians

\*\* To aid in understanding, we use the symbol “rad” in place of the number 1 (one) for this dimensionless derived unit.

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|                                  |                                                                                      |                                  |
|----------------------------------|--------------------------------------------------------------------------------------|----------------------------------|
| $C_{M_q} + C_{M_{\dot{\alpha}}}$ | Pitch damping moment coefficient                                                     | $1/\text{rad}$                   |
| $C_{N_q} + C_{N_{\dot{\alpha}}}$ | Pitch damping force coefficient                                                      | $1/\text{rad}$                   |
| $C_{spin}$                       | Spin damping moment coefficient                                                      | <i>none</i>                      |
| $d$                              | Reference diameter of projectile                                                     | $m$                              |
| $\overrightarrow{DF}$            | Drag force                                                                           | $\text{kg } m/s^2$               |
| $\vec{E}$                        | Position of the center of mass of the body with respect to spherical earth's surface | $m$                              |
| $\vec{F}$                        | Total force acting on the center of mass of the projectile or rocket                 | $\text{kg } m/s^2$               |
| $f_D$                            | Drag factor                                                                          | <i>none</i>                      |
| $f_L$                            | Lift Factor                                                                          | <i>none</i>                      |
| $f_T$                            | Thrust factor for rocket motor                                                       | <i>none</i>                      |
| $\overrightarrow{FCM}$           | Angular momentum due to fin cant moment                                              | $\text{kg } m^2 \text{ rad}/s^2$ |
| $\vec{g}$                        | Acceleration due to gravity                                                          | $m/s^2$                          |
| $g_0$                            | Magnitude of acceleration due to gravity at mean sea level                           | $m/s^2$                          |
| $\vec{H}$                        | Total angular momentum of the body                                                   | $\text{kg } m^2 \text{ rad}/s$   |
| $\dot{\vec{H}}$                  | Rate of change of the total angular momentum of the body                             | $\text{kg } m^2 \text{ rad}/s^2$ |
| $\vec{H}_{(t_L)}$                | Total angular momentum of the body at $t_L$                                          | $\text{kg } m^2 \text{ rad}/s$   |
| $I_{SP}$                         | Specific impulse of rocket motor fuel                                                | $N \text{ s}/\text{kg}$          |
| $I_x$                            | Axial moment of inertia                                                              | $\text{kg } m^2$                 |
| $I_{X(t_L)}$                     | Axial moment of inertia of the rocket at $(t_L)$                                     | $\text{kg } m^2$                 |

|                        |                                                                               |                    |
|------------------------|-------------------------------------------------------------------------------|--------------------|
| $I_Y$                  | Transversal moment of inertia of the body                                     | $kg\ m^2$          |
| $I_{Y(t_L)}$           | Transversal moment of inertia of the rocket at $(t_L)$                        | $kg\ m^2$          |
| $i$                    | Form factor                                                                   | <i>none</i>        |
| $l_W$                  | Distance from weapon trunnion to muzzle                                       | <i>m</i>           |
| $lat$                  | Latitude of launch point; for southern hemisphere $lat$ is negative           | <i>deg</i>         |
| $\overrightarrow{LF}$  | Lift force                                                                    | $kg\ m/s^2$        |
| $m$                    | Fuzed projectile or rocket mass at time $t$                                   | <i>kg</i>          |
| $\dot{m}$              | Fuzed projectile or rocket mass rate of change at time $t$                    | $kg/s$             |
| $\dot{m}_f$            | Mass flow rate of the motor fuel                                              | $kg/s$             |
| $m_r$                  | Reference fuzed projectile or rocket mass                                     | <i>kg</i>          |
| $M$                    | Local Mach number                                                             | <i>none</i>        |
| $\overrightarrow{MF}$  | Magnus Force                                                                  | $kg\ m/s^2$        |
| $\overrightarrow{MM}$  | Angular momentum due to Magnus moment                                         | $kg\ m^2\ rad/s^2$ |
| $\overrightarrow{OM}$  | Angular momentum due to overturning moment                                    | $kg\ m^2\ rad/s^2$ |
| $p$                    | Axial spin rate of projectile                                                 | <i>rad/s</i>       |
| $p_0$                  | Initial spin rate of projectile                                               | <i>rad/s</i>       |
| $p_{(t_L)}$            | Axial spin rate of rocket at $t_L$                                            | <i>rad/s</i>       |
| $P$                    | Local atmospheric air pressure                                                | <i>Pa</i>          |
| $P_r$                  | Standard atmospheric air pressure at sea level<br>( $1.01325 \times 10^5$ Pa) | <i>Pa</i>          |
| $\overrightarrow{PDF}$ | Pitch damping force                                                           | <i>N</i>           |

|                        |                                                                                                                                      |                     |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| $\overrightarrow{PDM}$ | Angular momentum due to pitch damping moment                                                                                         | $kg\ m^2\ rad/s^2$  |
| $Q_D$                  | Yaw drag fitting factor                                                                                                              | <i>none</i>         |
| $QE$                   | Quadrant elevation                                                                                                                   | <i>mil</i>          |
| $Q_M$                  | Magnus force fitting factor                                                                                                          | <i>none</i>         |
| $\vec{R}$              | Position of the center of the sphere, locally approximating the geoid                                                                | <i>m</i>            |
| $R$                    | Radius of the sphere, locally approximating the geoid                                                                                | <i>m</i>            |
| $\vec{r}$              | Position of the center of mass of the projectile or rocket with respect to the center of the sphere, locally approximating the geoid | <i>m</i>            |
| $r$                    | Distance of the center of mass of the projectile or rocket from the center of the sphere, locally approximating the geoid            | <i>m</i>            |
| $r_e$                  | Distance from the body center-of-mass to the motor nozzle exit                                                                       | <i>m</i>            |
| $r_{ne}$               | Radius of the motor's mass flow exit                                                                                                 | <i>m</i>            |
| $r_t$                  | Distance from the body center-of-mass to the motor nozzle throat                                                                     | <i>m</i>            |
| $\overrightarrow{SDM}$ | Angular momentum due to spin damping moment                                                                                          | $kg\ m^2\ rad/s^2$  |
| $t$                    | Time variable (computed time of flight)                                                                                              | <i>s</i>            |
| $t_c$                  | Twist of rifling at muzzle                                                                                                           | <i>calibers/rev</i> |
| $t_{FM}$               | Time of first rocket motion                                                                                                          | <i>s</i>            |
| $t_L$                  | Time at end of rocket launch phase                                                                                                   | <i>s</i>            |
| $t_{FO}$               | Time at rocket fin opening                                                                                                           | <i>s</i>            |
| $t_{E5D}$              | Time at end of 5DOF phase for rockets                                                                                                | <i>s</i>            |

|                         |                                                                                                         |                    |
|-------------------------|---------------------------------------------------------------------------------------------------------|--------------------|
| $\overrightarrow{TF}$   | Thrust force due to rocket motor thrust                                                                 | $N$                |
| $\overrightarrow{TJDM}$ | Angular momentum due to transverse jet damping moment                                                   | $kg\ m^2\ rad/s^2$ |
| $T_V$                   | Virtual temperature                                                                                     | $^{\circ}C$        |
| $t_0$                   | Time zero (beginning of calculations)                                                                   | $s$                |
| $\vec{u}$               | Velocity of the projectile or rocket with respect to ground-fixed axes                                  | $m/s$              |
| $\dot{\vec{u}}$         | Acceleration of center-of-mass in the fixed coordinate system                                           | $m/s^2$            |
| $u_0$                   | Initial speed of projectile or rocket with respect to ground-fixed axes                                 | $m/s$              |
| $v$                     | Speed of projectile or rocket with respect to air                                                       | $m/s$              |
| $\vec{v}$               | Velocity of the projectile or rocket with respect to air                                                | $m/s$              |
| $\vec{w}$               | Velocity of the air with respect to the ground (wind velocity)                                          | $m/s$              |
| $\vec{x}$               | Unit vector along longitudinal axis of the rigid body                                                   | <i>none</i>        |
| $\vec{x}_0$             | Unit vector along longitudinal axis of the rigid body, Initially                                        | <i>none</i>        |
| $\vec{x}_{(t_L)}$       | Unit vector along longitudinal axis of the rigid body, at $t_L$                                         | <i>none</i>        |
| $\dot{\vec{x}}$         | Rate of change of unit vector along longitudinal axis of the body                                       | $rad/s$            |
| $\vec{X}$               | Position vector of the center of mass of the projectile or rocket in the ground-fixed coordinate system | $m$                |

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|                   |                                                                                                                                               |                         |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| $\vec{X}_0$       | Initial position vector of the center of mass of the projectile or rocket                                                                     | <i>m</i>                |
| $X_{2w}$          | Height of weapon trunnion above mean sea level                                                                                                | <i>m</i>                |
| $\alpha$          | Total angle of attack                                                                                                                         | <i>rad</i>              |
| $\vec{\alpha}_e$  | Yaw of repose approximation                                                                                                                   | <i>rad</i>              |
| $\Delta AZ$       | Difference between the weapon azimuth (bearing) and the $\vec{1}$ axis measured clockwise                                                     | <i>mil</i>              |
| $\Delta t$        | Integration time step                                                                                                                         | <i>s</i>                |
| $\Delta \vec{w}$  | Difference in wind speed between the integration steps                                                                                        | <i>m/s</i>              |
| $\Delta \vec{u}$  | Velocity correction for windage jump                                                                                                          | <i>m/s</i>              |
| $\varepsilon$     | Fin cant angle                                                                                                                                | <i>rad</i>              |
| $\gamma$          | Ratio of specific heats of air, 1.4                                                                                                           | <i>none</i>             |
| $\vec{\Lambda}$   | Acceleration due to Coriolis effect                                                                                                           | <i>m/s<sup>2</sup></i>  |
| $\mu_f$           | Coefficient of sliding friction                                                                                                               | <i>none</i>             |
| $\rho$            | Density (specific mass) of air                                                                                                                | <i>kg/m<sup>3</sup></i> |
| $\psi$            | Orientation of the yaw                                                                                                                        | <i>rad</i>              |
| $\vec{\omega}$    | Angular velocity of the coordinate system due to the angular speed of the earth                                                               | <i>rad/s</i>            |
| $\omega_{y(t_L)}$ | Transverse angular velocity of the body at $(t_L)$ about an axis perpendicular to the quadrant elevation of the weapon in the vertical plane. | <i>rad/s</i>            |
| $\omega_{z(t_L)}$ | Transverse angular velocity of the body at $(t_L)$ in the vertical plane containing the weapon.                                               | <i>rad/s</i>            |
| $\Omega$          | Angular speed of the earth                                                                                                                    | <i>rad/s</i>            |
| •                 | Denotes differentiation with respect to time when appears over variable or function                                                           | <i>s<sup>-1</sup></i>   |
| →                 | Denotes vector when appears over variable                                                                                                     | <i>none</i>             |

## III. COMPARISON OF AERODYNAMIC COEFFICIENT SYMBOLS

Three different conventions have been widely used to describe the aerodynamic coefficients associated with the force and moment system used to describe the motion of the center of mass of a projectile. Table III-1 provides a concise comparison of the symbols. The first column lists the aerodynamic coefficient symbols used throughout this document. These symbols are associated with a drag force and lift force whose directions are defined with respect to the velocity vector of the center of mass of the shell and perpendicular to that vector (in the plane of yaw) respectively. The NACA symbols are associated with axial drag and normal forces whose directions are defined with respect to the axis of the shell and perpendicular to that axis (also in the plane of yaw) respectively. Consequently a rotational transformation is necessary to completely relate these two sets of symbols:

$$C_{D_0} = C_{x_0} \quad (57)$$

$$C_{D_{\alpha^2}} = C_{x_{\alpha^2}} + C_{z_{\alpha}} - \frac{1}{2}C_{x_0} \quad (58)$$

$$C_{L_{\alpha}} = C_{z_{\alpha}} - C_{x_0} \quad (59)$$

$$C_{L_{\alpha^3}} = C_{z_{\alpha^3}} - \frac{1}{2}C_{z_{\alpha}} - C_{x_{\alpha^2}} \quad (60)$$

The ballistic symbols shown have been slightly modernized to account for the fact that the original set of symbols did not allow for some of the dependence of the aerodynamic forces on yaw angle. The same logic as that used by the aeronautical community was applied to affix the subscripts  $\theta$ ,  $\alpha$ ,  $\alpha^2$  and  $\alpha^3$  to the original symbols.

NOTE: Equations (58) and (60) use the small angle assumption  $\cos \alpha = (1 - \sin^2 \alpha)^{1/2}$  approximately equal to  $1 - \frac{1}{2}\alpha^2$ . If instead one assumes  $\cos \alpha$  is approximately equal to 1, then (58) and (60) become  $C_{D_{\alpha^2}} = C_{x_{\alpha^2}} + C_{z_{\alpha}}$  and  $C_{L_{\alpha^3}} = C_{z_{\alpha^3}}$ , respectively. If one uses higher order coefficients, such as  $C_{D_{\alpha^4}}$  and  $C_{L_{\alpha^5}}$ , (58) and (60) may still be used with a negligible effect on precision. The axial drag force and the drag force,  $\overrightarrow{DF}$ , used in this STANAG are assumed to act in the same direction.



**Table III-1. Aerodynamic Coefficient Symbols**

| Symbol                             | Other Symbols                      |                    | C-to-K Relationship                                     | Definition                            |
|------------------------------------|------------------------------------|--------------------|---------------------------------------------------------|---------------------------------------|
|                                    | NACA                               | Ballistic          |                                                         |                                       |
| $C_{D_0}$                          | $C_{x_0}$                          | $K_{D_0}$          | $K_{D_0} = \frac{\pi}{8} C_{D_0}$                       | Drag force coefficient                |
| $C_{D_{\alpha^2}}$                 | $C_{x_{\alpha^2}}$                 | $K_{D_{\alpha^2}}$ | $K_{D_{\alpha^2}} = \frac{\pi}{8} C_{D_{\alpha^2}}$     | Quadratic yaw drag force coefficient  |
| $C_{D_{\alpha^4}}$                 | $C_{x_{\alpha^4}}$                 | $K_{D_{\alpha^4}}$ | $K_{D_{\alpha^4}} = \frac{\pi}{8} C_{D_{\alpha^4}}$     | Quartic yaw drag force coefficient    |
| $C_{L_{\alpha}}$                   | $-C_{z_{\alpha}}$                  | $K_{L_{\alpha}}$   | $K_{L_{\alpha}} = \frac{\pi}{8} C_{L_{\alpha}}$         | Lift force coefficient                |
| $C_{L_{\alpha^3}}$                 | $C_{z_{\alpha^3}}$                 | $K_{L_{\alpha^3}}$ | $K_{L_{\alpha^3}} = \frac{\pi}{8} C_{L_{\alpha^3}}$     | Cubic lift force coefficient          |
| $C_{L_{\alpha^5}}$                 | $C_{z_{\alpha^5}}$                 | $K_{L_{\alpha^5}}$ | $K_{L_{\alpha^5}} = \frac{\pi}{8} C_{L_{\alpha^5}}$     | Quintic lift force coefficient        |
| $C_{M_{\alpha}}$                   | $C_{M_{\alpha}}$                   | $K_{M_{\alpha}}$   | $K_{M_{\alpha}} = \frac{\pi}{8} C_{M_{\alpha}}$         | Overturning moment coefficient        |
| $C_{M_{\alpha^3}}$                 | $C_{M_{\alpha^3}}$                 | $K_{M_{\alpha^3}}$ | $K_{M_{\alpha^3}} = \frac{\pi}{8} C_{M_{\alpha^3}}$     | Cubic over-Turning moment coefficient |
| $C_{mag-f}$                        | $C_{y_{p\alpha}}$                  | $K_F$              | $K_F = -\frac{\pi}{8} C_{mag-f}$                        | Magnus force coefficient              |
| $C_{mag-m}$                        | $C_{n_{p\alpha}}$                  | $K_T$              | $K_T = -\frac{\pi}{8} C_{mag-m}$                        | Magnus moment coefficient             |
| $(C_{N_q} + C_{N_{\dot{\alpha}}})$ | $(C_{Z_q} + C_{Z_{\dot{\alpha}}})$ | $K_S$              | $K_S = -\frac{\pi}{8} (C_{N_q} + C_{N_{\dot{\alpha}}})$ | Damping force coefficient             |
| $(C_{M_q} + C_{M_{\dot{\alpha}}})$ | $(C_{m_q} + C_{m_{\dot{\alpha}}})$ | $K_H$              | $K_H = -\frac{\pi}{8} (C_{M_q} + C_{M_{\dot{\alpha}}})$ | Damping moment coefficient            |
| $C_{spin}$                         | $C_{l_p}$                          | $K_A$              | $K_A = -\frac{\pi}{8} C_{spin}$                         | Spin damping moment coefficient       |
| $C_{l_{\delta}}$                   | <i>none</i>                        | $K_E$              | $K_E = \frac{\pi}{8} C_{l_{\delta}}$                    | Fin cant coefficient                  |

NOTE: Special symbols  $C_{mag-f}$  and  $C_{spin}$  are defined in place of  $C_{Np\alpha}$  and  $C_{lp}$  to avoid possible confusion where these symbols have been used differently in published data (notably in the UK and US), where:

$$C_{mag-f} = C_{Np\alpha} \quad \text{for data nondimensionalized using } pd/v \quad (61)$$

$$= (1/2) C_{yp\alpha} \quad \text{for data nondimensionalized using } pd/2v \quad (62)$$

$$C_{mag-m} = C_{Mp\alpha} \quad \text{for data nondimensionalized using } pd/v \quad (63)$$

$$= (1/2) C_{np\alpha} \quad \text{for data nondimensionalized using } pd/2v \quad (64)$$

$$C_{spin} = C_{lp} \quad \text{for data nondimensionalized using } pd/v \quad (65)$$

$$= (1/2) C_{lp} \quad \text{for data nondimensionalized using } pd/2v \quad (66)$$

#### IV. LIST OF DATA REQUIREMENTS

| <b><i>Physical Data Requirements</i></b>                             | <b><i>Fin-Stabilized</i></b> | <b><i>Spin-Stabilized</i></b> |
|----------------------------------------------------------------------|------------------------------|-------------------------------|
| Initial speed of projectile with respect to ground (muzzle velocity) | $u_0$                        | $u_0$                         |
| Twist of rifling at muzzle                                           |                              | $t_c$                         |
| Initial mass of fuze projectile                                      | $m_0$                        | $m_0$                         |
| Reference mass of fuze projectile                                    | $m_r$                        | $m_r$                         |
| Reference diameter                                                   | $d$                          | $d$                           |
| Initial axial moment of inertia                                      | $I_{x_0}$                    | $I_{x_0}$                     |
| Axial spin rate of rocket at $(t_L)$                                 | $P_{(t_L)}$                  |                               |
| Axial moment of inertia at $(t_L)$                                   | $I_{X_{(t_L)}}$              |                               |
| Initial transverse moment of inertia                                 | $I_{Y_0}$                    |                               |
| Transverse moment of inertia at $(t_L)$                              | $I_{Y_{(t_L)}}$              |                               |
| Specific impulse of rocket fuel                                      | $I_{SP}$                     |                               |
| Mass flow rate of the motor fuel                                     | $\dot{m}_f$                  |                               |
| Coefficient of friction                                              | $\mu_f$                      |                               |
| Fin cant angle                                                       | $\varepsilon$                |                               |
| Distance from body center of mass to motor nozzle exit               | $r_e$                        |                               |
| Radius of motors mass flow exit                                      | $r_{ne}$                     |                               |
| Distance from body center of mass to motor nozzle throat             | $r_t$                        |                               |

| <b><i>Aerodynamic Data Requirements</i></b> | <b><i>Fin-Stabilized</i></b>     | <b><i>Spin-Stabilized</i></b> |
|---------------------------------------------|----------------------------------|-------------------------------|
| Zero yaw drag force coefficient             | $C_{D_0}$                        | $C_{D_0}$                     |
| Quadratic yaw drag force coefficient        | $C_{D_{\alpha^2}}$               | $C_{D_{\alpha^2}}$            |
| Quartic yaw drag force coefficient          |                                  | $C_{D_{\alpha^4}}$            |
| Total drag force coefficient                | $C_D$                            |                               |
| Lift force coefficient                      | $C_{L_{\alpha}}$                 | $C_{L_{\alpha}}$              |
| Cubic lift force coefficient                | $C_{L_{\alpha^3}}$               | $C_{L_{\alpha^3}}$            |
| Quintic lift force coefficient              |                                  | $C_{L_{\alpha^5}}$            |
| Magnus force coefficient                    | $C_{mag-f}$                      | $C_{mag-f}$                   |
| Overturning moment coefficient              | $C_{M_{\alpha}}$                 | $C_{M_{\alpha}}$              |
| Cubic overturning moment coefficient        | $C_{M_{\alpha^3}}$               | $C_{M_{\alpha^3}}$            |
| Spin damping moment coefficient             | $C_{spin}$                       | $C_{spin}$                    |
| Magnus moment coefficient                   | $C_{mag-m}$                      |                               |
| Pitch damping moment coefficient            | $C_{M_q} + C_{M_{\dot{\alpha}}}$ |                               |
| Pitch damping force coefficient             | $C_{N_q} + C_{N_{\dot{\alpha}}}$ |                               |
| Fin cant moment coefficient                 | $C_{l_{\delta}}$                 |                               |

NOTE: Aerodynamic coefficients for fin-stabilized rockets are required for both the fin closed and fin open configurations.

| <b><i>Fitting Factor Data Requirement</i></b>                                                          | <b><i>Fin-Stabilized</i></b> | <b><i>Spin-Stabilized</i></b> |
|--------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------|
| <b><i>Quadrant Elevation Fitting</i></b>                                                               |                              |                               |
| Form Factor                                                                                            | $i$                          | $i$                           |
| Ballistic Coefficient                                                                                  |                              | $C$                           |
| <b><i>Mach Number Fitting</i></b>                                                                      |                              |                               |
| Thrust Factor                                                                                          | $f_T$                        |                               |
| Drag Factor                                                                                            |                              | $f_D$                         |
| <b><i>Quadrant Elevation and Mach Number Fitting</i></b>                                               |                              |                               |
| Adjustment to azimuth – difference between actual and computed deflection – for fin-stabilized rockets | $\Delta(\Delta AZ)$          |                               |
| Lift Factor                                                                                            | $f_L$                        | $f_L$                         |
| Yaw Drag Factor                                                                                        |                              | $Q_D$                         |
| Magnus Factor                                                                                          |                              | $Q_M$                         |
| Exit area of motor jet                                                                                 | $A_e$                        |                               |
| Transverse vertical angular velocity of the body at ( $t_L$ )                                          | $\omega_{y(t_L)}$            |                               |
| Transverse horizontal angular velocity of the body at ( $t_L$ )                                        | $\omega_{z(t_L)}$            |                               |
| Adjustment to time-of-flight – difference between actual and computed time-of-flight                   | $\Delta ToF$                 | $\Delta ToF$                  |

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IMPLEMENTATION OF THE AGREEMENT

24. This STANAG is implemented when a nation has issued instructions to the agencies concerned to use the Modified Point Mass Trajectory Model for spin-stabilized projectiles and a Five Degrees of Freedom Model for exterior ballistic trajectory simulation of fin-stabilized rockets as detailed in this agreement.

**FORMS OF FIRE CONTROL INPUT DATA**

## 1. Aerodynamic coefficients

Aerodynamic coefficients are given as a function of Mach number. These functions are in the form of polynomials of fourth degree or less. They are defined over regions of Mach number, from  $M_{MAX_{i-1}}$  up to and including  $M_{MAX_i}$ .

Each aerodynamic coefficient is described by a series of polynomials of the form:

$$C_i = a_0 + a_1M + a_2M^2 + a_3M^3 + a_4M^4 \quad (\text{A.1})$$

$C_i$  is a particular aerodynamic coefficient  
 $M$  is Mach number

## 2. Fitting Factor:

## a. Functions of Quadrant Elevation

Form factor ( $i$ ) or ballistic coefficient ( $C$ ) and lift factor ( $f_L$ ) are given for each charge as follows:

$$f = a_0 + a_1(QE) + a_2(QE)^2 + a_3(QE)^3 \quad (\text{A.2})$$

where  $f = i, C$ , or  $f_L$

Yaw drag factor ( $Q_D$ ) and Magnus force factor ( $Q_M$ ) have been chosen as constants. (See Table A-1)

## b. Functions of Mach Number:

Drag factor ( $f_D$ ) and lift factor ( $f_L$ ) are quartic polynomials of Mach number. Yaw drag factor ( $Q_D = 1.2$ ) has been chosen as a constant and Magnus force factor ( $Q_M = 1$ ). (See Table A-1)

## c. Function of Computed Time

The modeling of time used in this document may require a correction to the computed time of flight for each charge as follows:

$$T = t + a_0 + a_1t + a_2t^2 + a_3t^3 \quad (\text{A.3})$$

$T$  is time of flight

$t$  is computed time of flight

FACTORS IN EQUATIONS FOR MATCHING OBSERVED RANGE  
FIRING DATA

Two fitting systems are used, quadrant elevation or Mach number as shown below:

**Table A-1.** Basic Fitting Data

| Fitting to                              | Fitting Data as a Function of                        |                                                   |
|-----------------------------------------|------------------------------------------------------|---------------------------------------------------|
|                                         | Quadrant Elevation<br>[One function for each charge] | Mach Number<br>[Same function for all charges]    |
| First Order<br>Range, all<br>elevations | *Form factor: $i$<br><br>(Drag factor: $f_D = 1$ )   | Drag factor: $f_D$<br><br>(Form factor: $i = 1$ ) |
| Drift                                   | Lift factor: $f_L$                                   | Lift factor: $f_L$                                |
| Second Order<br>Range, high<br>angle    | Yaw drag factor: $Q_D$                               | Yaw drag factor: $Q_D$                            |
| Vertex height<br>and time of<br>flight  | Magnus force factor: $Q_M$                           | Magnus force factor: $Q_M$                        |

\*In many countries a ballistic coefficient,  $C = m_r / i d^2$ , is in use as a fitting factor instead of the nondimensional form factor  $i$ .

To compensate for the approximations in the equations of motion, aerodynamic coefficients, rocket, projectile, rocket-assisted motor, and base burn motor performance, the factors contained in Table A-2 are applied in order to create correspondence between the computed and the observed range testing results.

**Table A-2.** Fitting factors for matching observed range firing data

| Range Firing Data                                           | Fitting Factors                            |                                            |                                       |                                       |
|-------------------------------------------------------------|--------------------------------------------|--------------------------------------------|---------------------------------------|---------------------------------------|
|                                                             | 5 DoF                                      | Modified Point Mass                        |                                       |                                       |
|                                                             | Fin-Stabilized Rockets                     | Projectiles                                | Rocket-Assisted Projectiles           | Base-Burn Projectiles                 |
| Motor Burn Time                                             | $f_{BT_p} \equiv 0$<br>$f_{BT_p} \equiv 0$ | $f_{BT_p} \equiv 0$<br>$f_{BT_p} \equiv 0$ | $f_{BT_p}^*$<br>$f_{BT_p} \equiv 0$   | $f_{BT_p}$<br>$f_{BT_p}$              |
| Velocity After Rocket Burning                               | $f_T$                                      | $f_T \equiv 0$                             | $f_T$                                 | $f_T \equiv 0$                        |
| Change in Radial Velocity During Rocket Burning             | $A_e$                                      | $A_e \equiv 0$                             | $A_e$                                 | $A_e \equiv 0$                        |
| Velocity Vertical and Horizontal Angle after Rocket Burning | $\omega_{y(t_L)}$<br>$\omega_{z(t_L)}$     |                                            |                                       |                                       |
| Range                                                       | $i$<br>$f(i_{BB,MT}) \equiv 0$             | $i$<br>$f(i_{BB,MT}) \equiv 0$             | $i$<br>$f(i_{BB,MT}) \equiv 0$        | $i$<br>$f(i_{BB,MT})$                 |
| Deflection                                                  | $\Delta(\Delta AZ)$<br>$f_L \equiv 1$      | $\Delta(\Delta AZ) \equiv 0$<br>$f_L$      | $\Delta(\Delta AZ) \equiv 0$<br>$f_L$ | $\Delta(\Delta AZ) \equiv 0$<br>$f_L$ |
| Time-of-Flight                                              | $\Delta ToF$                               | $\Delta ToF$                               | $\Delta ToF$                          | $\Delta ToF$                          |

\*Optional

NOTE: See Annex C or D for further fitting factors for base-burn and rocket-assisted projectiles



### 3. Fuze Setting

Fuze setting is given as a function of time of flight and initial spin as follows:

$$FS = a_0 + a_1T + a_2p_0 + a_3Tp_0 + a_4Tp_0^2 + a_5p_0^2 \quad (A.4)$$

or,

$$FS = \frac{T + a_0 + a_2p_0}{b_0 + b_1p_0} \quad (A.5)$$

$FS$  is fuze setting

$T$  is time of flight

$p_0$  is initial axial spin of projectile

### 4. Muzzle Velocity Corrections

#### a. Muzzle Velocity Correction for Propellant Temperature

Muzzle velocity correction for propellant temperature is given for each charge as follows:

$$\Delta U_{pt} = a_0 + a_1(PT - 21) + a_2(PT - 21)^2 + a_3(PT - 21)^3 \quad (A.6)$$

$\Delta U_{pt}$  is incremental change in muzzle velocity

$PT$  is propellant temperature

#### b. Muzzle Velocity Correction for Cannon (Tube) Wear

The muzzle velocity correction for cannon (tube) wear for a specific charge/projectile/cannon combination is given by

$$\Delta U_w = \frac{\Delta U_r(a_0 + a_1(EFC) + a_2(EFC)^2)}{a_{0,i} + a_{1,i}(EFC) + a_{2,i}(EFC)^2 + a_{3,i}(EFC)^3} \quad (A.7)$$

$\Delta U_w$  is incremental change in muzzle velocity

$\Delta U_r(EFC)$  is incremental change in muzzle velocity for the reference charge, i.e., the maximum charge compatible with the projectile/cannon combination evaluated at the current EFC count.

$EFC$  is the current number of Equivalent Full Charge firings from the cannon.

$a_{0,i}, a_{1,i}, a_{2,i}$  are the coefficients required to calculate the ratio of the incremental change in muzzle velocity for the reference change ( $\Delta U_r$ ) to the corresponding incremental

change in muzzle velocity for the  $i^{th}$  charge at the specified EFC. This ratio is derived from the interior ballistics models or actual firing trials.

c. Muzzle Velocity Correction for Projectile Mass

Muzzle velocity correction for projectile mass is given for each charge as follows:

$$\Delta U_m = n_i U_i (m - m_r) / m_r \quad (\text{A.8})$$

$\Delta U_m$  is incremental change in muzzle velocity

$n_i$  is a constant factor derived from interior ballistic models or projectile mass firings

$U_i$  is firing table muzzle velocity for the  $i^{th}$  charge corrected for calibration,  
propellant temperature and cannon wear

$m$  is fuzed projectile mass

$m_r$  is reference mass of fuzed projectile

5. List of Additional Symbols for Forms of Fire Control Data

| <u>Symbol</u>   | <u>Definition</u>                                                   | <u>Units</u>       |
|-----------------|---------------------------------------------------------------------|--------------------|
| $a_i$           | Constant coefficient                                                | <i>none</i>        |
| $b_i$           | Constant coefficient                                                | <i>none</i>        |
| $FS$            | Fuze setting                                                        | <i>none</i>        |
| $n$             | Constant coefficient                                                | <i>none</i>        |
| $PT$            | Propellant temperature                                              | $^{\circ}\text{C}$ |
| $T$             | Time of flight                                                      | <i>s</i>           |
| $U$             | Firing table muzzle velocity                                        | <i>m/s</i>         |
| $\Delta U_m$    | Change in muzzle velocity due to nonstandard Projectile mass        | <i>m/s</i>         |
| $\Delta U_{pt}$ | Change in muzzle velocity due to nonstandard Propellant temperature | <i>m/s</i>         |

|                     |                                                                                                        |            |
|---------------------|--------------------------------------------------------------------------------------------------------|------------|
| $\Delta U_w$        | Change in muzzle velocity due to cannon (tube) wear                                                    | <i>m/s</i> |
| $\Delta(\Delta AZ)$ | Adjustment to azimuth – difference between actual and computed deflection – for fin-stabilized rockets | <i>mil</i> |

**ADDITIONAL TERMS FOR PROJECTILES WITH BOURRELET NUBS**

The presence of nubs introduces terms in the expressions for certain of the aerodynamic forces and moments, which are proportional to the nub setting angle and independent of the spin rate.

**1. Equations of Motion**

Define  $C_{mag-f}$  and  $C_{spin}$  as follows:

$$C_{mag-f} = C_{N_{pa}} + C_{N_{\delta_n}} \delta_n / (pd / v) \quad \text{for data nondimensionalized using } pd / v \quad (\text{B.1})$$

$$= [C_{y_{pa}} + C_{y_{\delta_n}} \delta_n / (pd / 2v)] / 2 \quad \text{for data nondimensionalized using } pd / 2v \quad (\text{B.2})$$

$$C_{spin} = C_{l_p} + C_{l_{\delta_n}} \delta_n / (pd / v) \quad \text{for data nondimensionalized using } pd / v \quad (\text{B.3})$$

$$= [C_{l_p} + C_{l_{\delta_n}} \delta_n / (pd / 2v)] / 2 \quad \text{for data nondimensionalized using } pd / 2v \quad (\text{B.4})$$

**2. List of Additional Symbols**

| <u>Symbol</u>                            | <u>Definition</u>                                                                                                                                                                | <u>Units</u>   |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| $C_{l_{\delta_n}}$                       | Spin moment coefficient, due to canted bourrelet nubs, at zero spin                                                                                                              | $1/\text{rad}$ |
| $C_{N_{\delta_n}}$ or $C_{y_{\delta_n}}$ | Side force coefficient, due to canted bourrelet nubs, at zero spin                                                                                                               | $1/\text{rad}$ |
| $\delta_n$                               | Orientation of nub principal axis with respect to a plane containing the projectile axis and the nub geometrical center. Positive $\delta_n$ produces a positive rolling moment. | $\text{rad}$   |

**3. Additional Data Requirements**

| <b><i>Physical Data</i></b> | <b><i>Symbol</i></b> |
|-----------------------------|----------------------|
| Nub orientation             | $\delta_n$           |

| <b><i>Aerodynamic Data</i></b>       | <b><i>Symbol</i></b>                     |
|--------------------------------------|------------------------------------------|
| Spin moment coefficient at zero spin | $C_{l_{\delta_n}}$                       |
| Side force coefficient at zero spin  | $C_{N_{\delta_n}}$ or $C_{y_{\delta_n}}$ |

**ADDITIONAL TERMS FOR SPIN-STABILIZED ROCKET-ASSISTED  
PROJECTILES, SPIN-STABILIZED BASE-BURN PROJECTILES, AND FIN-  
STABILIZED ROCKETS – METHOD 1**

1. Introduction

This annex provides the equations required to simulate the flight of rocket-assisted projectiles and the specific equations required to simulate the flight of base-burn projectiles and fin-stabilized rockets.

2. Additional Terms for Spin-Stabilized Rocket-Assisted Projectiles

- a. Add the following thrust term to the equation of motion of the center of mass of the projectile shown at Eq. (1):

$$\frac{\overrightarrow{TF}}{m} = \frac{T^*}{m} \left( \frac{\vec{v} \cos \alpha_e}{v} + \vec{\alpha}_e \right) \quad (C.1)$$

- b. Zero yaw drag coefficient during the burning phase ( $t_{DI} \leq t \leq t_B$ ) is  $C_{D_{oT}}$ .  
c. Thrust during the burning phase ( $t_{DI} \leq t \leq t_B$ ) is as follows:

$$T^* = f_T T_R + (P_r - P) A_e, \quad (C.2)$$

where:

$$T_R = T_{ST} (t_{DI_{ST}} - t_{B_{ST}}) / (t_{DI} - t_B), \quad (C.3)$$

and

$$t^* = (t - t_{DI}) [(t_{DI_{ST}} - t_{B_{ST}}) / (t_{DI} - t_B)] + t_{DI_{ST}} \quad (C.4)$$

- d. Mass of the projectile is given by:

$$\text{mass at } t = 0 \text{ is } m = m_0 \quad (C.5)$$

for  $t < t_{DI}$

$$\dot{m} = -m_{DI} / t_{DI} \quad (C.6)$$

$$\text{mass at } t = t_{DI} \text{ is } m = m_0 - m_{DI} - m_{DOB} \quad (C.7)$$

for  $t_{DI} \leq t < t_B$

$$\dot{m} = -T_R / I_{SP} \quad (C.8)$$

for  $t \geq t_B$

$$\dot{m} = 0 \quad (C.9)$$

$$m = m_B \quad (C.10)$$

where:

$$m_B = m_0 - m_{DI} - m_{DOB} - m_f \quad (C.11)$$

### 3. Additional Terms for Spin-Stabilized Base-Burn Projectiles

- a. The change in acceleration due to the base drag reduction of a base-burn motor,  $\overrightarrow{BB}$  during burning ( $t_{DI} \leq t$  and  $m \geq m_B$ ) is added to the equation of motion of the center of mass of the projectile which follows from Equation (1) of the main body of this STANAG:

$$\overrightarrow{BB} = + \left[ \frac{\left( \frac{\pi}{8} \right) \rho d^2 v^2 C_{x_{BB}} f(I) f(i_{BB}, MT)}{m} \right] \left( \frac{\vec{v} \cos \alpha_e}{v} + \vec{\alpha}_e \right) \quad (C.12)$$

- b. The coefficient  $C_{x_{BB}}$  is the drag reduction coefficient during the burning phase. As with other aerodynamic coefficients, values of this coefficient are given by polynomial functions of Mach number of fourth degree or less.
- c. The characteristic flow rate function of the base-burn motor is given as follows:

$$f(I) = I/I_0 \quad \text{If } I \leq I_0 \quad (C.13)$$

$$f(I) = 1 \quad \text{If } I \geq I_0 \quad (C.14)$$

where:  $I$  is the base-burn motor fuel injection parameter;

$$I = \frac{4 \dot{m}_f}{\pi d_b^2 \rho v}, \quad (C.15)$$

and  $I_0$  is the injection parameter permitting optimum efficiency of the base-burn motor. It is given as a function of Mach number. These functions are in the form of polynomials of one degree or less. They are defined over regions of Mach number, from  $M_{MAX_{i-1}}$  up to and including  $M_{MAX_i}$ .

NOTE: The function  $f(I) = I/I_0$  is principally used for emptying phase and eventually at the start of the burning phase.

- d. The coefficient  $i_{BB}$  is a fitting factor which can be used, if necessary, to adjust the drag reduction.  $i_{BB}$  is expressed as a polynomial function of the quadrant elevation,  $QE$ , of third degree or less. It is included for matching observed firing data during the base-burn phase.
- e. Mass of the projectile is given by:

for  $0 \leq t < t_{DI}$

$$m = m_0 - m_{CB_0} \quad (C.16)$$

for  $t_{DI} \leq t$  and  $m \geq m_B$

$$\dot{m} = \dot{m}_f \quad (C.17)$$

$$\dot{m}_f = -V_C \rho_p S_C (m_{CB}) \quad (C.18)$$

where:

$$m_B = m_0 - m_f \quad (C.19)$$

$$m_{CB} = m_0 - m \quad (C.20)$$

$m_{CB_0}$  is the mass of fuel burnt in the barrel;

$V_C$  is the combustion rate;

$\rho_p$  is the density of fuel;

$S_C (m_{CB})$  is the area of combustion at time  $t$  and will be expressed in the form of a function of the mass of fuel burnt:

$$S_C = a_i + b_i m_{CB}; \quad (C.21)$$

for  $m_{CB_i} < m_{CB} \leq m_{CB_{i+1}}$

$a_i$  and  $b_i$  are defined over regions of  $m_{CB}$ , from  $m_{CB_i=0}$  up to and including  $m_{CB_i=n}$ .

The combustion rate is given by:

$$V_C = V_{C_0} f(MT) g(P) K(p) \quad (C.22)$$

where:

$V_{C_0}$  is the combustion rate obtained on the strand burner at standard pressure and temperature

$MT$  is the motor fuel temperature

$$f(MT) = e^{\beta(MT-21)} \quad (C.23)$$

$P$  is the local atmospheric air pressure

$$g(P) = k P^n \quad (C.24)$$

$p$  is the axial spin of the projectile

$K(p)$  is determined from experiments to take into account the influence of axial spin on the combustion rate,  $K(p)$  is a linear function of spin for each charge.

The time of motor burnout,  $t_B$ , is the time for which  $m = m_B$  and it is a program output.



#### 4. Fitting Factors

To compensate for the approximations in this annex, certain fitting factors are applied in order to create correspondence between the computed and the observed range testing results.

**Table C-1.** Fitting Factors for Rocket-Assisted and Base Burn Projectiles

| Fitting                                        | Fitting Function            |                        |
|------------------------------------------------|-----------------------------|------------------------|
|                                                | Rocket-assisted Projectiles | Base-burn Projectiles  |
| Change in Radial Velocity During Motor Burning | $f_T$                       |                        |
| Base-burn, Motor-burn Time                     | $t_{DI}$<br>$t_B - t_{DI}$  | $t_{DI}$<br>$K(p)$     |
| Range                                          | $i$                         | $i$<br>$f(i_{BB, MT})$ |

5. The location of the center of mass for base burn and rocket-assisted projectiles is given by:

$$X_{CG} = X_{CG_0} + \left[ \frac{(X_{CG_0} - X_{CG_B})(m - m_0)}{m_0 - m_B} \right] \quad (C.25)$$

6. Common Equations for base burn and rocket-assisted, spin-stabilized projectiles, and fin-stabilized rockets.

- a. The overturning moment coefficient of the munition is given by:

$$C_{M_\alpha} = C_{M_\alpha}^* + \left[ \frac{(X_{CG} - X_{CG_0})(C_{D_0} + C_{L_\alpha})}{d} \right] \quad (C.26)$$

where:  $C_{M_\alpha}^*$  is determined for the initial munition configuration and, when  $t_{DI} \leq t < t_B$ , then  $C_{D_0}$  equals  $C_{D_{0T}}$  for rocket-assisted projectiles and fin-stabilized rockets.

- b. The cubic overturning moment coefficient of the munition is given by:

$$C_{M_{\alpha^3}} = C_{M_{\alpha^3}}^* + \left[ \frac{(X_{CG} - X_{CG_0})(C_{L_{\alpha^3}} + C_{D_{\alpha^2}} - 1/2 C_{L_\alpha})}{d} \right] \quad (C.27)$$

where:  $C_{M_{\alpha^3}}^*$  is determined for the initial munition configuration.

- c. The axial moment of inertia of the munition is given by:

$$I_x = I_{x_0} + \left[ \frac{(I_{x_0} - I_{x_B})(m - m_0)}{m_0 - m_B} \right] \quad (C.28)$$

7. The location of the center of mass, motor nozzle exit, motor nozzle throat, and transverse moment of inertia for fin-stabilized rockets.

- a. The location of the center of mass is given by:

$$X_{CG} = X_{CG_B} + \left[ \frac{(X_{CG_{f_0}} - X_{CG_B})(m - m_B)}{m} \right] \quad (C.29)$$

where:

$$X_{CG_{f_0}} = X_{CG_B} + \frac{(X_{CG_0} - X_{CG_B})m_0}{(m_0 - m_B)}$$

- b. The location of the motor nozzle exit from the center of the mass is given by:

$$r_e = \ell - X_{CG} \quad (C.30)$$

- c. The location of the motor nozzle throat from the center of mass is given by:

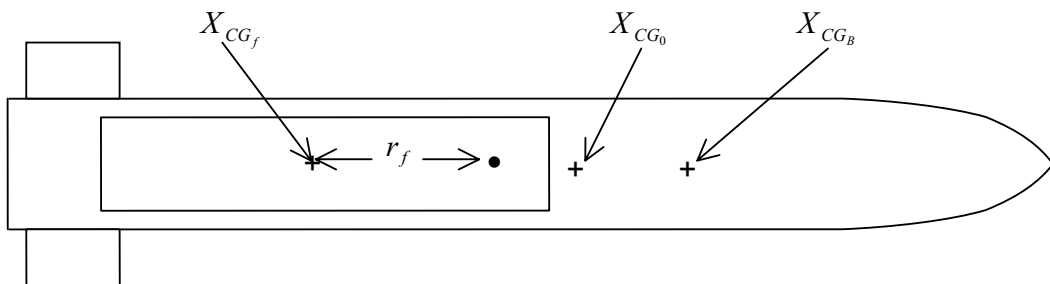
$$r_t = r_e - r_{t-t} \quad (C.31)$$

- d. The transverse moment of inertia of a fin-stabilized rocket is given by:

$$I_Y = I_{Y_0} - (m_0 - m)r_f^2 + m_0 \left( X_{CG_{f_0}} - X_{CG_0} \right)^2 - m \left( X_{CG_f} - X_{CG} \right)^2 \quad (C.32)$$

where:  $r_f$  is the radius of gyration of the fuel mass.

Figure C-1 illustrates the distances used for the determination of the transverse moment of inertia of the rocket or projectile during motor burning.



**Figure C-1.** Distances Used for the Determination of the Transverse Moment of Inertia

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8. List of Additional Symbols for Rocket-Assisted and Base-Burn Projectiles

| <u>Symbol</u>         | <u>Definition</u>                                                               | <u>Units</u> |
|-----------------------|---------------------------------------------------------------------------------|--------------|
| $A_e$                 | Exit area of jet                                                                | $m^2$        |
| $a_i$                 | Coefficient (constant)                                                          | <i>none</i>  |
| $b_i$                 | Coefficient (constant)                                                          | <i>none</i>  |
| $\overrightarrow{BB}$ | Acceleration due to drag reduction of base-burn motor                           | $m/s^2$      |
| $C_{D_{0T}}$          | Zero yaw drag coefficient (thrust on)                                           | <i>none</i>  |
| $C_{M_\alpha}^*$      | Overturning moment coefficient for initial projectile configuration             | <i>none</i>  |
| $C_{x_{bb}}$          | Drag reduction coefficient during base-burn motor burning                       | <i>none</i>  |
| $d$                   | Reference diameter of projectile                                                | $m$          |
| $d_b$                 | Diameter of projectile base                                                     | $m$          |
| $e$                   | Base of natural logarithms                                                      | <i>none</i>  |
| $f(i_{BB, MT})$       | Base-burn factor                                                                | <i>none</i>  |
| $f(I)$                | Function $I$                                                                    | <i>none</i>  |
| $f(MT)$               | Combustion rate as a function of motor fuel temperature                         | <i>none</i>  |
| $f_T$                 | Thrust Factor                                                                   | <i>none</i>  |
| $g(P)$                | Combustion rate as a function of atmospheric air pressure                       | <i>none</i>  |
| $i_{BB}$              | Fitting factor to adjust the drag reduction as a function of quadrant elevation | <i>none</i>  |
| $I$                   | Base-burn motor fuel injection parameter                                        | <i>none</i>  |
| $I_0$                 | Base-burn motor fuel injection parameter for optimum efficiency                 | <i>none</i>  |

|             |                                                                |             |
|-------------|----------------------------------------------------------------|-------------|
| $I_{SP}$    | Specific impulse                                               | $Ns/kg$     |
| $I_x$       | Axial moment of inertia of the projectile                      | $kg\ m^2$   |
| $I_{x0}$    | Initial axial moment of inertia                                | $kg\ m^2$   |
| $I_{xB}$    | Axial moment of inertia at burnout                             | $kg\ m^2$   |
| $I_y$       | Transverse moment of inertia                                   | $kg\ m^2$   |
| $I_{y0}$    | Initial transverse moment of inertia                           | $kg\ m^2$   |
| $k$         | Constant in burning rate versus pressure formula               | <i>none</i> |
| $K(p)$      | Axial spin burning rate factor                                 | <i>none</i> |
| $\ell$      | Distance of the motor nozzle exit from nose                    | $m$         |
| $m_0$       | Initial fuzed projectile mass                                  | $kg$        |
| $m_B$       | Fuzed projectile mass at burnout                               | $kg$        |
| $m_{CB}$    | Mass of motor fuel burnt                                       | $kg$        |
| $m_{CB0}$   | Mass of motor fuel burnt in the barrel                         | $kg$        |
| $m_{DI}$    | Ignition delay element mass                                    | $kg$        |
| $m_{DOB}$   | Delay obturator mass                                           | $kg$        |
| $m_f$       | Projectile fuel mass                                           | $kg$        |
| $\dot{m}_f$ | Mass flow rate of the motor fuel                               | $kg/s$      |
| $MT$        | Temperature of motor fuel                                      | $^{\circ}C$ |
| $n$         | Exponent in burning rate versus pressure formula               | <i>none</i> |
| $P$         | Air pressure                                                   | $Pa$        |
| $P_r$       | Reference air pressure for standard thrust                     | $Pa$        |
| $r_e$       | Distance from the body center-of-mass to the motor nozzle exit | $m$         |

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|                |                                                                          |          |
|----------------|--------------------------------------------------------------------------|----------|
| $r_f$          | Radius of gyration of the motor fuel mass                                | $m$      |
| $r_t$          | Distance from the body center-of-mass to the motor nozzle throat         | $m$      |
| $r_{t-t}$      | Distance of the motor nozzle exit from the motor nozzle throat           | $m$      |
| $S_C$          | Area of combustion at time $t$                                           | $m^2$    |
| $t$            | Computed time of flight                                                  | $s$      |
| $t^*$          | Pseudo-computed time for mapping of thrust at nonstandard conditions     | $s$      |
| $t_B$          | Time of rocket motor burnout                                             | $s$      |
| $t_{BST}$      | Standard time of rocket motor burnout                                    | $s$      |
| $t_{DI}$       | Time of rocket motor ignition delay                                      | $s$      |
| $t_{DIST}$     | Standard time of rocket motor ignition delay                             | $s$      |
| $T_R$          | Thrust produced by rocket motor at time $t$                              | $N$      |
| $T_{ST}$       | Standard thrust as function of burning time                              | $N$      |
| $T^*$          | Effective thrust                                                         | $N$      |
| $V_C$          | Combustion rate of base-burn fuel                                        | $m/s$    |
| $V_{C_0}$      | Combustion rate of base-burn fuel on strand burner                       | $m/s$    |
| $X_{CG}$       | Distance of center of mass from nose at time $t$                         | $m$      |
| $X_{CG_0}$     | Initial distance of center of mass from nose                             | $m$      |
| $X_{CG_B}$     | Distance of center of mass from nose at burnout                          | $m$      |
| $X_{CG_{f_0}}$ | Distance of center-of-mass of the rocket motor fuel from nose, initially | $m$      |
| $\beta$        | Base-burn motor temperature fuel burning coefficient                     | $none$   |
| $\rho_p$       | Density of base-burn motor fuel                                          | $kg/m^3$ |

## 9. Additional Data Requirements for Rocket-Assisted and Base-Burn Projectiles

| <b>(1) Physical Data</b>                                       | <b>Symbol</b> |
|----------------------------------------------------------------|---------------|
| Exit area of jet                                               | $A_e$         |
| Reference diameter of projectile                               | $d$           |
| Axial moment of inertia of the projectile                      | $I_x$         |
| Initial axial moment of inertia of the projectile              | $I_{x0}$      |
| Axial moment of inertia of the projectile at motor burnout     | $I_{xB}$      |
| Transverse moment of inertia of the rocket, initially          | $I_{Y0}$      |
| Distance of the motor nozzle exit from nose                    | $\ell$        |
| Fuzed projectile mass at burnout                               | $m_B$         |
| Initial mass of fuzed projectile                               | $m_0$         |
| Radius of gyration of motor fuel mass                          | $r_f$         |
| Distance of the motor nozzle exit from the motor nozzle throat | $r_{t-t}$     |
| Initial distance of center of mass from nose                   | $X_{CG0}$     |
| Distance of center of mass from nose at burnout                | $X_{CGB}$     |

| <b>(2) Aerodynamic Data</b>                                 | <b>Symbol</b>   |
|-------------------------------------------------------------|-----------------|
| Zero Yaw drag coefficient (thrust on)                       | $C_{D0T}$       |
| Overturning moment coefficient for initial fuzed projectile | $C_{M\alpha}^*$ |
| Drag reduction coefficient during base-burn motor burning   | $C_{xBB}$       |

| <b>(3) Motor Data – Rocket-Assisted</b>      | <b>Symbol</b> |
|----------------------------------------------|---------------|
| Specific impulse                             | $I_{SP}$      |
| Ignition delay element mass                  | $m_{DI}$      |
| Delay obturator mass                         | $m_{dob}$     |
| Mass of rocket motor fuel                    | $m_f$         |
| Reference air pressure for standard thrust   | $P_r$         |
| Time of rocket motor burnout                 | $t_B$         |
| Standard time of motor burnout               | $t_{BST}$     |
| Time of rocket motor ignition delay          | $t_{DI}$      |
| Standard time of rocket motor ignition delay | $t_{DI_{ST}}$ |
| Standard thrust as function of pseudo-time   | $T_{ST}$      |

| <b>(4) Motor Data – Base-Burn</b>                                                                                                                                                                                                                                          | <b>Symbol</b> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Base-burn motor fuel injection parameter for optimum efficiency                                                                                                                                                                                                            | $I_0$         |
| Mass of fuel burnt in the barrel                                                                                                                                                                                                                                           | $m_{CB_0}$    |
| Mass of motor fuel                                                                                                                                                                                                                                                         | $m_f$         |
| Area of combustion expression in the form of a function of the mass of fuel burnt<br>$S_C = a_i + b_i m_{CB}$ ; for $m_{CB_i} < m_{CB} \leq m_{CB_{i+1}}$<br>$a_i$ and $b_i$ are defined over regions of $m_{CB}$ , from $m_{CB_{i=0}}$ up to and including $m_{CB_{i=n}}$ | $S_C$         |
| Combustion rate of base-burn fuel on strand burner                                                                                                                                                                                                                         | $V_{C_0}$     |
| Base-burn motor temperature fuel burning coefficient                                                                                                                                                                                                                       | $\beta$       |
| Exponent in burning rate versus pressure formula                                                                                                                                                                                                                           | $n$           |
| Density of base-burn motor fuel                                                                                                                                                                                                                                            | $\rho_p$      |
| Constant in burning rate versus pressure formula                                                                                                                                                                                                                           | $k$           |

## 10. Fire Control Data for Rocket-Assisted and Base-Burn Motors

### a. Rocket-Assisted Motor

Time of rocket ignition delay ( $t_{DI}$ ), time of rocket motor burn ( $t_B - t_{DI}$ ) and thrust factor ( $f_T$ ) are given as a function of rocket motor temperature for each charge as follows:

$$f = a_0 + a_1(MT - 21) + a_2(MT - 21)^2 + a_3(MT - 21)^3 \quad (C.33)$$

where:

$$f = t_{DI}, t_B - t_{DI}, \text{ or } f_T \quad (C.34)$$

$MT$  is rocket motor temperature

Standard Thrust is given as a function of pseudo-computed time ( $t^*$ ) as follows:

$$T_{ST} = a_0 + a_1 t^* + a_2 t^{*2} + a_3 t^{*3} \quad (C.35)$$



b. Base-Burn Motor

The axial spin influence on burning rate factor  $K(p)$ , is given as a linear function of spin for each charge.

$I_0$  is a linear function of Mach number.

The factor,  $f(i_{BB, MT})$ , for a base-burn projectile is given as a function of quadrant elevation,  $QE$ , and motor temperature,  $MT$ , for each charge as follows:

$$i_{BB(MT=21)} = a_0 + a_1 QE + a_2 QE^2 + a_3 QE^3 \quad (C.36)$$

and

$$f(i_{BB, MT}) = i_{BB(MT=21)} + b_1(MT - 21) + b_2(MT - 21)^2 + b_3(MT - 21)^3 + b_4(MT - 21)^4 \quad (C.37)$$

Time of base-burn ignition delay ( $t_{DI}$ ) is given as a function of motor temperature for each charge.

$$t_{DI} = a_0 + a_1(MT - 21) + a_2(MT - 21)^2 + a_3(MT - 21)^3 \quad (C.38)$$

$MT$  is rocket motor temperature

**ADDITIONAL TERMS FOR SPIN-STABILIZED ROCKET-ASSISTED PROJECTILES,  
SPIN-STABILIZED BASE-BURN PROJECTILES, AND FIN-STABILIZED ROCKETS  
– METHOD 2**

1. Introduction

This annex provides the equations required to simulate the flight of spin-stabilized rocket-assisted projectiles, spin-stabilized base-burn projectiles, and fin-stabilized rockets

2. Equations for spin-stabilized rocket-assisted projectiles, spin-stabilized base-burn projectiles, and fin-stabilized rockets

a. For rocket-assisted projectiles:

The acceleration due to thrust force of the rocket motor,  $\frac{\overrightarrow{TF}}{m}$ , during burning ( $t_{DI} \leq t \leq t_B$ ) is added to the equation of motion of the center of mass of the unassisted projectile:

$$\frac{\overrightarrow{TF}}{m} = \left[ \frac{f_T \dot{m}_f I_{SP} + (P_r - P) A_e}{m} \right] \left( \frac{\vec{v} \cos \alpha_e}{v} + \vec{\alpha}_e \right) \quad (D.1)$$

During rocket motor burning the aerodynamic zero-yaw coefficient is  $C_{DoT}$  and if  $\dot{m}_f \leq \dot{m}_p$  then  $(P_r - P) A_e = 0$ .

b. For base-burn projectiles:

The base drag reduction due to a base-burn motor during burning ( $t_{DI} \leq t \leq t_B$ ) is added to the drag term  $\left( \frac{\overrightarrow{DF}}{m} \right)$  of the projectile.

$$\frac{\overrightarrow{DF}}{m} = - \frac{\pi \rho d^2 i}{8m} \{ C_{Do} - f(i_{BB,MT}) \} \left[ \frac{I \left( \frac{\delta BP}{\delta I} \right)}{\left( \frac{\gamma}{2} \right) M^2 \left( \frac{d}{d_b} \right)^2} + C_{D_{a^2}} (Q_D \alpha_e)^2 + C_{D_{a^4}} (Q_D \alpha_e)^4 \right] v \vec{v} \quad (D.2)$$

where:

$$I = \frac{4 \dot{m}_f}{\pi \rho v d_b^2} \quad (D.3)$$

$$f(i_{BB,MT}) \left[ \frac{\left( \frac{\dot{m}_f}{\rho v A_b} \right) \left( \frac{\delta BP}{\delta I} \right)}{\left( \frac{\gamma}{2} \right) M^2 \left( \frac{d}{d_b} \right)^2} \right]$$

is used to represent the drag reduction due to the mass flow ( $\dot{m}_f$ ) of the base-burn motor. The factor  $f(i_{BB,MT})$  is included for matching observed range firing data.

c. For fin-stabilized rockets:

See Eq. (16) and (25) in main document.

### 3. Common Equations for base-burn and rocket-assisted, spin-stabilized projectiles, and fin-stabilized rockets.

a. The mass flow for spin-stabilized projectiles and fin-stabilized rockets is given by:

at  $t = 0$

$$m = m_0 \quad (D.4)$$

$$\dot{m} = 0 \quad (D.5)$$

for  $0 < t < t_{DI}$

$$\dot{m} = -\frac{m_{DI}}{t_{DI}} \quad (D.6)$$

$$m = -\int_0^{t_{DI}} \left( \frac{m_{DI}}{t_{DI}} \right) dt + m_0 \quad (D.7)$$

for  $t_{DI} \leq t < t_B$

$$\dot{m} = -\dot{m}_f \quad (D.8)$$

$$m = -\int_{t_{DI}}^{t_B} \dot{m}_f dt - \int_0^{t_{DI}} \left( \frac{m_{DI}}{t_{DI}} \right) dt + m_0 \quad (D.9)$$

$$\dot{m}_f = \left( \frac{t_B^* - t_{(t)}^*}{t_{B(t)} - t} \right) \dot{m}_f^* \quad (D.10)$$

$$\dot{t}_{(t)}^* = \frac{t_B^* - t_{(t)}^*}{t_{B(t)} - t} \quad (D.11)$$

$$\dot{t}_{B(t)} = (t_{B(t)} - t) \left[ f_{BT_p} \left( \frac{\dot{P}}{P} \right) + f_{BT_p} \left( \frac{\dot{P}}{P} \right) \right] \quad (D.12)$$

where:

$$\dot{P} = \frac{\Delta P}{\Delta t} = \frac{\Delta P}{\Delta E_2} u_2 \quad (D.13)$$

and at  $t_{DI}$

$$t_{(t_{DI})}^* = t_{DI}^* \quad (D.14)$$

$$t_{B(t_{DI})} = t_{DI} + \left[ (t_B - t_{DI}) \left( \frac{P_{(t_{DI})}}{P_r} \right)^{f_{BT_p}} \left( \frac{P_{(t_{DI})}}{P_r} \right)^{f_{BT_p}} \right] \quad (D.15)$$

(Time-of-motor burnout as a function of  $t_{DI}$ )

|                |   |                                                  |
|----------------|---|--------------------------------------------------|
| $p_r$          | = | Reference axial spin rate for motor mass flow    |
| $p_{(t_{DI})}$ | = | Actual spin rate for motor mass flow at $t_{DI}$ |
| $P_r$          | = | Standard atmospheric air pressure                |
| $P_{(t_{DI})}$ | = | Actual atmospheric air pressure at $t_{DI}$      |

for  $t \geq t_B$

$$m_B = m_0 - m_{DI} - m_f \quad (D.16)$$

$$\dot{m} = 0 \quad (D.17)$$

b. The location of the center of mass of the munition is given by:

$$X_{CG} = X_{CG_B} + \left[ \frac{(X_{CG_{f_0}} - X_{CG_B})(m - m_B)}{m} \right] \quad (D.18)$$

where:

$$X_{CG_{f_0}} = X_{CG_B} + \frac{(X_{CG_0} - X_{CG_B})m_0}{(m_0 - m_B)} \quad (D.19)$$

- c. The location of the motor nozzle exit from the center of the mass is given by:

$$r_e = \ell - X_{CG} \quad (D.20)$$

- d. The location of the motor nozzle throat from the center of mass is given by:

$$r_t = r_e - r_{t-t} \quad (D.21)$$

- e. The overturning moment coefficient of the munition is given by:

$$C_{M_\alpha} = C_{M_\alpha}^* + \left[ \frac{(X_{CG} - X_{CG_0})(C_{D_0} + C_{L_\alpha})}{d} \right] \quad (D.22)$$

where:  $C_{M_\alpha}^*$  is determined for the initial munition configuration and, when  $t_{DI} \leq t < t_B$ , then  $C_{D_0}$  equals  $C_{D_{0T}}$  for rocket-assisted projectiles and fin-stabilized rockets. The aerodynamic coefficients are considered to be dimensionless, with unit = 1, in this formulation.

- f. The cubic overturning moment coefficient of the munition is given by:

$$C_{M_{\alpha^3}} = C_{M_{\alpha^3}}^* + \left[ \frac{(X_{CG} - X_{CG_0})(C_{L_{\alpha^3}} + C_{D_{\alpha^2}} - 1/2 C_{L_\alpha})}{d} \right] \quad (D.23)$$

where:  $C_{M_{\alpha^3}}^*$  is determined for the initial munition configuration. The aerodynamic coefficients are considered to be dimensionless, with unit =1, in this formulation.

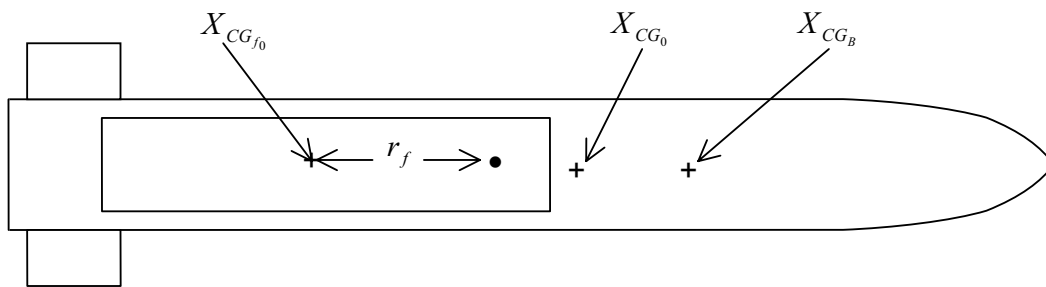
- g. The axial moment of inertia of the munition is given by:

$$I_x = I_{x_0} + \left[ \frac{(I_{x_0} - I_{x_B})(m - m_0)}{m_0 - m_B} \right] \quad (D.24)$$

4. The transverse moment of inertia of a fin-stabilized rocket is given by:

$$I_Y = I_{Y_0} - (m_0 - m) r_f^2 + m_0 (X_{CG_{f_0}} - X_{CG_0})^2 - m (X_{CG_f} - X_{CG})^2 \quad (D.25)$$

Figure D-1 illustrates the distances used for the determination of the transverse moment of inertia of the rocket during motor burning.



**Figure D-1.** Distances Used for the Determination of the Transverse Moment of Inertia

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5. List of Additional Symbols

| <u>Symbol</u>      | <u>Definition</u>                                                               | <u>Units</u> |
|--------------------|---------------------------------------------------------------------------------|--------------|
| $A_b$              | Area of projectile base                                                         | $m^2$        |
| $A_e$              | Exit area of the motor jet                                                      | $m^2$        |
| $C_{D_0}$          | Zero yaw drag coefficient                                                       | <i>none</i>  |
| $C_{D_{0T}}$       | Zero yaw drag coefficient during rocket motor burning                           | <i>none</i>  |
| $C_{M_\alpha}^*$   | Overturning moment coefficient for initial fuzed munition                       | <i>none</i>  |
| $C_{M_\alpha^3}^*$ | Cubic overturning moment coefficient for initial fuzed munition                 | $1/rad^2$    |
| $d_b$              | Diameter of projectile base                                                     | $m$          |
| $f_T$              | Thrust factor                                                                   | <i>none</i>  |
| $f(i_{BB}, MT)$    | Base-burn factor                                                                | <i>none</i>  |
| $f_{BT_p}$         | Base-burn motor spin rate burning-time factor                                   | <i>none</i>  |
| $f_{BT_P}$         | Base-burn motor atmospheric air pressure burning-time factor                    | <i>none</i>  |
| $I$                | Base-burn motor fuel injection parameter                                        | <i>none</i>  |
| $i_{BB}$           | Fitting factor to adjust the drag reduction as a function of quadrant elevation | <i>none</i>  |
| $I_x$              | Axial moment of inertia of the munition at time $t$                             | $kg\ m^2$    |
| $I_{x_0}$          | Axial moment of inertia of the munition, initially                              | $kg\ m^2$    |
| $I_{x_B}$          | Axial moment of inertia of the munition at motor burnout                        | $kg\ m^2$    |
| $I_Y$              | Transverse moment of inertia of the rocket at time $t$                          | $kg\ m^2$    |
| $I_{Y_0}$          | Transverse moment of inertia of the rocket, initially                           | $kg\ m^2$    |
| $\ell$             | Distance of the motor nozzle exit from nose                                     | $m$          |

|               |                                                                                                |             |
|---------------|------------------------------------------------------------------------------------------------|-------------|
| $m_0$         | Fuzed munition mass, initially                                                                 | $kg$        |
| $m_B$         | Fuzed munition mass at burnout                                                                 | $kg$        |
| $m_{DI}$      | Mass of ignition delay element                                                                 | $kg$        |
| $m_f$         | Mass of motor fuel                                                                             | $kg$        |
| $\dot{m}_f^*$ | Reference mass flow rate of the motor fuel as a function of $t^*$ pseudo-time-of-motor burning | $kg/s$      |
| $\dot{m}_p$   | Minimum mass flow rate of the motor fuel for air pressure term                                 | $kg/s$      |
| $MT$          | Temperature of motor fuel                                                                      | $^{\circ}C$ |
| $p$           | Axial spin rate of projectile                                                                  | $rad/s$     |
| $p_r$         | Reference axial spin rate for motor mass flow                                                  | $rad/s$     |
| $r_e$         | Distance from the body center-of-mass to the motor nozzle exit                                 | $m$         |
| $r_f$         | Radius of gyration of the motor fuel mass                                                      | $m$         |
| $r_t$         | Distance from the body center-of-mass to the motor nozzle throat                               | $m$         |
| $r_{t-t}$     | Distance of the motor nozzle exit from the motor nozzle throat                                 | $m$         |
| $t^*$         | Pseudo-time-of-motor burning                                                                   | $s$         |
| $t_B$         | Time-of-motor burnout                                                                          | $s$         |
| $t_B^*$       | Reference time-of-motor burnout                                                                | $s$         |
| $t_{DI}$      | Time-of-motor ignition delay                                                                   | $s$         |
| $t_{DI}^*$    | Reference time-of-motor ignition delay                                                         | $s$         |
| $X_{CG}$      | Distance of center of mass of the munition from nose at time $t$                               | $m$         |



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|                              |                                                                                                 |             |
|------------------------------|-------------------------------------------------------------------------------------------------|-------------|
| $X_{CG_0}$                   | Distance of center of mass of the munition from nose at time $t_0$                              | $m$         |
| $X_{CG_B}$                   | Distance of center of mass of the munition from nose at time $t_B$                              | $m$         |
| $X_{CG_{f_0}}$               | Distance of center-of-mass of the rocket motor fuel from nose at time $t_0$                     | $m$         |
| $\frac{\delta BP}{\delta I}$ | Change in non-dimensional base pressure for a change in the base-burn motor injection parameter | <i>none</i> |
| $\frac{\Delta P}{\Delta t}$  | Rate of change of atmospheric air pressure as seen by the munition                              | $Pa/s$      |

6. Additional Data Requirements

| <b><i>Physical data</i></b>                                           | <b><i>Symbol</i></b> |
|-----------------------------------------------------------------------|----------------------|
| Area of projectile base                                               | $A_b$                |
| Exit area of motor jet                                                | $A_e$                |
| Diameter of projectile base                                           | $d_b$                |
| Axial moment of inertia of the munition, initially                    | $I_{x_0}$            |
| Axial moment of inertia of the munition at motor burnout              | $I_{x_B}$            |
| Transverse moment of inertia of the rocket, initially                 | $I_{Y_0}$            |
| Distance of the motor nozzle exit from nose                           | $\ell$               |
| Fuzed munition mass, initially                                        | $m_0$                |
| Fuzed munition mass at burnout                                        | $m_B$                |
| Mass of ignition delay element                                        | $m_{DI}$             |
| Mass of motor fuel                                                    | $m_f$                |
| Minimum mass flow rate of the fuel for air pressure term              | $\dot{m}_p$          |
| Radius of gyration of motor fuel mass                                 | $r_f$                |
| Distance of the motor nozzle exit from the motor nozzle throat        | $r_{t-t}$            |
| Distance of center of mass of the munition from nose, initially       | $X_{CG_0}$           |
| Distance of center of mass of the munition from nose at motor burnout | $X_{CG_B}$           |

| <b><i>Aerodynamic data</i></b>                                      | <b><i>Symbol</i></b> |
|---------------------------------------------------------------------|----------------------|
| Zero yaw drag coefficient during rocket motor burning               | $C_{D_{0T}}$         |
| Overturning moment coefficient for initial fuzed munition           | $C_{M_\alpha}^*$     |
| Nonlinear overturning moment coefficient for initial fuzed munition | $C_{M_{\alpha^3}}^*$ |

| <b>Motor data</b>                                                                                                                                                                                                                                                                        | <b>Symbol</b>                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Reference mass flow rate of the motor fuel as a function of pseudo-time-of motor burning, $t^*$<br>$\dot{m}_f^* = a_0 + a_1 t^* + a_2 t^{*2}$ ; for $t_i^* < t^* \leq t_{i+1}^*$<br>$a_0$ and $a_1$ are defined over regions of $t^*$ , from $t_{i=0}^*$ up to and including $t_{i=n}^*$ | $\dot{m}_f^*$                |
| Reference axial spin rate for motor mass flow                                                                                                                                                                                                                                            | $p_r$                        |
| Standard atmospheric air pressure at sea level                                                                                                                                                                                                                                           | $P_r$                        |
| Time-of-motor burnout                                                                                                                                                                                                                                                                    | $t_B$                        |
| Reference time-of-motor burnout                                                                                                                                                                                                                                                          | $t_B^*$                      |
| Time-of-motor ignition delay                                                                                                                                                                                                                                                             | $t_{DI}$                     |
| Reference time-of-motor ignition delay                                                                                                                                                                                                                                                   | $t_{DI}^*$                   |
| Specific impulse of motor fuel                                                                                                                                                                                                                                                           | $I_{SP}$                     |
| Change in non-dimensional base pressure for a change in the base-burn motor injection parameter as a function of Mach number and injection parameter                                                                                                                                     | $\frac{\delta BP}{\delta I}$ |

$\frac{\delta BP}{\delta I}$  is given, for up to five values of  $I$ , as a function of Mach number in the form of polynomials of fourth degree or less. They are defined over regions of Mach number, from  $M_{MAX_i=1}$  up to and including  $M_{MAX}$ .

$$\left( \frac{\delta BP}{\delta I} \right)_I = a_0 + a_1 M + a_2 M^2 + a_3 M^3 + a_4 M^4 \quad (D.26)$$

## 7. Fire Control Data for Projectile Motor

(for a complete fitting factor table, see Annex A)

- a. The time-of-motor ignition delay,  $t_{DI}$ , and time-of-motor burn,  $(t_B - t_{DI})$ , for rocket-assisted and base-burn projectiles are given as a function of motor temperature,  $MT$ , for each charge as follows:

$$t_{DI} = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.27)$$

$$(t_B - t_{DI}) = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.28)$$

- b. The thrust factor,  $f_T$ , for a rocket-assisted projectile is given as a function of motor temperature,  $MT$ , for each charge as follows:

$$f_T = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.29)$$

- c. The spin rate burning-time factor,  $f_{BT_p}$ , for a base-burn projectile is given as a constant for each charge.
- d. The atmospheric air pressure burning-time factor,  $f_{BT_p}$ , for a base-burn projectile is given as a constant for each charge.
- e. The factor,  $f(i_{BB,MT})$ , for a base-burn projectile is given as a function of quadrant elevation,  $QE$ , and motor temperature,  $MT$ , for each charge as follows:

$$i_{BB(MT=21)} = a_0 + a_1 QE + a_2 QE^2 + a_3 QE^3 \quad (D.30)$$

and

$$f(i_{BB,MT}) = i_{BB(MT=21)} + b_1 (MT - 21) + b_2 (MT - 21)^2 + b_3 (MT - 21)^3 + b_4 (MT - 21)^4 \quad (D.31)$$

## 8. Fire Control Data for Fin Stabilized Rockets

(for a complete fitting factor table, see Annex A)

- a. The mass of ignition delay element,  $m_{DI}$ , time of motor ignition delay,  $t_{DI}$ , spin rate burning-time factor,  $f_{BT_p}$ , and the atmospheric air pressure burning-time factor,  $f_{BT_p}$ , are zero.
- b. The time of first motion,  $t_{FM}$ , time of launch,  $t_L$ , time of rocket fin opening,  $t_{FO}$ , time of motor burn,  $(t_B - t_{DI})$ , is given by,  $t_B$ , time of end of five degrees of freedom,  $t_{ESD}$ , and thrust factor,  $f_T$ , are given as a function of motor temperature,  $MT$ , as follows:

$$t_{FM} = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.32)$$

$$t_L = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.33)$$

$$t_{FO} = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.34)$$

$$t_B = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.35)$$

$$t_{ESD} = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.36)$$

$$f_T = a_0 + a_1 (MT - 21) + a_2 (MT - 21)^2 + a_3 (MT - 21)^3 \quad (D.37)$$

- c. At time,  $t_L$ , the initial angular velocity (spin) of the fin stabilized rocket is,  $p_{(t_L)}$ , a constant and the initial transverse angular velocities of the fin stabilized rocket,  $\omega_{y_{(t_L)}}$  and  $\omega_{z_{(t_L)}}$ , are given as a function of quadrant elevation,  $QE$ , as follows:

$$p_{(t_L)} = a_0 \quad (D.38)$$

$$\omega_{y_{(t_L (MT=21))}} = a_0 + a_1 QE + a_2 QE^2 + a_3 QE^3 \quad (D.39)$$

and  $\omega_{y_{(t_L)}} = \omega_{y_{(t_L (MT=21))}} + b_1 (MT - 21) + b_2 (MT - 21)^2 + b_3 (MT - 21)^3$

$$\omega_{z_{(t_L (MT=21))}} = a_0 + a_1 QE + a_2 QE^2 + a_3 QE^3 \quad (D.40)$$

and  $\omega_{z_{(t_L)}} = \omega_{z_{(t_L (MT=21))}} + b_1 (MT - 21) + b_2 (MT - 21)^2 + b_3 (MT - 21)^3$

- d. The form factor,  $i$ , and adjustment to azimuth,  $\Delta(\Delta AZ)$ , are given as a function of quadrant elevation,  $QE$ , as follows:

$$i = a_0 + a_1 QE + a_2 QE^2 + a_3 QE^3 \quad (D.41)$$

$$\Delta(\Delta AZ) = a_0 + a_1 QE + a_2 QE^2 + a_3 QE^3 \quad (D.42)$$

- e. The modeling of time used for fin stabilized rockets may require a correction to time of flight as follows:

$$T = t + \Delta ToF \quad (D.43)$$

$$\Delta ToF = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \quad (D.44)$$

**CALCULATION OF SUBMUNITION TRAJECTORIES****1. Introduction**

This annex provides the methods for calculating submunition trajectories.

Submunition trajectories are the trajectories for submunitions ejected from the carrier shell in flight.

- a. Submunition trajectories may represent trajectories for:
  - (1) Cargo bomblets
  - (2) Mines
  - (3) Smoke canisters
  - (4) Illuminating canisters
  - (5) Canister for sensor-fuzed submunition
- b. This Annex describes methods for the calculation of:
  - (1) the base separation point
  - (2) the submunition trajectories with a point mass trajectory model
  - (3) simple trajectory calculation using a closed form solution
  - (4) submunition dispersion
  - (5) empty carrier trajectory

Figure E-1 defines the nomenclature for the different phases. A collective name for the different phases is *submunition trajectory*. Note that *terminal phase* should always be used for the phase in which the submunition hits the ground or functions.

**2. Rules for the Calculation of the Base Separation Point**

The base separation point is the point where the fuze of the carrier shell activates the first phase of the submunition trajectories. Calculation of the position of the base separation point can be done using either of the following rules:

- a. a predefined *height of burst (HOB)* above the target, or by
- b. requiring a certain *time of fall (T<sub>fs</sub>)* for the center submunition.

Both can be given either as a constant or as a polynomial function of the superelevation. Thus for rule (a.) and rule (b.) we respectively get

$$HOB = a_0 + a_1 (QE - A_S) + a_2 (QE - A_S)^2 + a_3 (QE - A_S)^3 + (a_4 + a_5 (QE - A_S) + a_6 (QE - A_S)^2 + a_7 (QE - A_S)^3) Alt_w \quad (E.1)$$

and

$$T_{fs} = b_0 + b_1(QE - A_s) + b_2(QE - A_s)^2 + b_3(QE - A_s)^3 \quad (E.2)$$

Rule (a.) is designed to ensure that a specified fraction of all base separation points occur at a certain height. Rule (b.) is designed to ensure that a specified fraction of submunitions hits the ground before the self-destruct time.

The starting velocity of any phase, which includes point-mass and closed-form-biases:

$$\vec{u}_{s1} = \vec{u}_{carrier} + \Delta u_{s1} \vec{c} \quad (E.3)$$

where:  $\vec{c} = \frac{\vec{v}}{v} \cos \alpha_e + \vec{\alpha}_e$  is the axis of the carrier shell at the end of the previous phase, and the vector  $\vec{\alpha}_e$  is the yaw of repose.

$\Delta u_{sn} = a_{0,n} + a_{1,n} p_{n-1}$  is the ejection speed of the submunition,  
and  $p_{n-1}$  is the magnitude of the spin at the end of the previous phase

For a more detailed description of ejection velocity from a spin-stabilized projectile used to estimate the maximum dimension of submunition pattern on the ground, see equation E.34.

NOTE: See Figure E-2 for explanation of additional terms

### 3. Calculation of Submunition Trajectories with a Point Mass Model

NOTE: See Annex F for Point Mass Model implementation.

The acceleration due to aerodynamic drag,  $\vec{D}$ , is computed as

$$\vec{D}_{sn} = -i_{sn} C_{Dsn} \frac{\rho S_{sn}}{2m_{sn}} v_{sn} \vec{v}_{sn} \quad (E.4)$$

where: the subscripts  $s$  and  $n$  represent submunition and the phase number respectively.  $S_{sn}$  is the effective presented cross sectional area of the submunition unit and  $m_{sn}$  is its mass. They can both be given as linear functions of time.

The wind velocity,  $\vec{w}$ , is accounted for through

$$\vec{v}_{sn} = \vec{u}_{sn} - \vec{w} \quad (E.5)$$

If necessary the spin damping can be calculated according to the expression

$$\dot{p}_{sn} = \frac{2\rho}{\pi I_{x_{sn}}} S_{sn}^2 p_{sn} C_{spin_{sn}} v_{sn} \quad (E.6)$$

Drag coefficient can be formatted in two ways, one of which is dependent on Mach number only and one of which is dependent on Mach number, velocity, and spin.

Drag coefficient is given as a function of the Mach number ( $M_{sn}$ ) of the submunition. The function is in the form of a linear function. It is defined over regions of Mach number similar to that of Annex A, paragraph 1. Thus in each interval

$$C_{D_{sn}} = a_{0,n} + a_{1,n} M_{sn} \quad (E.7)$$

$C_{D_{sn}}$  value depends on the Mach number and the quotient of velocity and spin as follows.

$$C_{D_{sn}} = \frac{1.0}{i_{sn}} + [a_{0,n} + a_{1,n} M_{sn}] \left( \frac{S_{ball\_sn} - S_{sn}}{S_{sn}} \right) \quad (E.8)$$

where:

$$S_{ball\_sn} = \frac{\pi}{4} d_{sn}^2 \quad (E.9)$$

and

$$d_{sn} = a_{0,sn} + \frac{a_{1,sn}}{1 + a_{2,sn} \left( \frac{|p_{sn}|}{u_{sn}} \right)^{a_{3,sn}}} \quad (E.10)$$

where: the coefficients  $a_{k,sn}$  ( $k=0,1,2,3$ ) are constant fitting factors for the submunition  $s$  in the phase  $n$ .

$p_{sn}$                       The spin of the submunition  $s$  in the phase  $n$  ( rad/s )  
 $u_{sn}$                       The velocity of the submunition  $s$  in the phase  $n$  ( m/s )

The form factor,  $i_{sn}$ , is given as a polynomial function of HOB

$$i_{sn} = b_0 + b_1 (HOB_{sn}) + b_2 (HOB_{sn})^2 + b_3 (HOB_{sn})^3 \quad (E.11)$$

Mass of submunition unit,  $m_{sn}$ , is given as a quadratic polynomial of  $t$

$$m_{sn} = a_0 + a_1 t + a_2 t^2 \quad (E.12)$$

Reference cross sectional area,  $S_n$ , is given as a quadratic polynomial function of  $t$

$$S_n = a_0 + a_1 t + a_2 t^2 \quad (E.13)$$

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and the corrected time is given by

$$T_{sn} = t_{sn} + a t_{sn} \quad (\text{E.14})$$

for each phase.

The end of the phase is determined either by height above the ground and/or the duration of the phase and is calculated by one of the following equations:

$$H_{phase} = a \quad (\text{E.15})$$

or

$$T_{f,sn} = a_0 + a_1(QE - A_S) + a_2(QE - A_S)^2 + a_3(QE - A_S)^3 + \frac{b_1}{u_{sn}} \quad (\text{E.16})$$

$u_{sn}$  is the velocity of the submunition at the beginning of the phase.

#### 4. Calculation Using Simplified Formulas

The ballistic trajectory of a phase can be calculated in closed form either by

##### a. Closed Form A

assuming a drag proportional to a velocity reference and constant meteorological parameters throughout the phase using the vector equation

$$\vec{x}_{n+1} = \vec{x}_n + \left( \frac{\vec{v}_{sn}}{k_s} - \frac{\vec{g}}{k_s^2} \right) [1 - \exp(-T_{sn} k_s)] + T_{sn} \left( \vec{w} + \frac{\vec{g}}{k_s} \right) \quad (\text{E.17})$$

where:

$$k_s = C_{D_{sn}} \rho \frac{S_{sn}}{2m_{sn}} v_{sn} \quad (\text{E.18})$$

$$\vec{g} = \begin{bmatrix} 0 \\ -g_0 \\ 0 \end{bmatrix} \quad (\text{E.19})$$

and

$$\vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} \quad (\text{E.20})$$

or



b. Closed Form B

assuming a constant velocity of fall and constant wind throughout the phase where the heights of burst and time of fall are related by

$$T_{f_{sn}} = \frac{HOB_n - HOB_{n+1}}{|v_{sn,2}|} \quad (E.21)$$

where  $v_{sn,2}$  is the vertical component of  $\vec{v}_{sn}$ .  $HOB_n$  is the height of burst at the start of phase  $n$ . If  $n$  is the last phase then  $HOB_{n+1} = 0$  meaning that the submunition terminates at ground level.

The displacement due to wind is  $T_{f_{sn}} \vec{w}_h$  where  $\vec{w}_h$  is the horizontal wind vector.

If  $HOB_n$  and  $HOB_{n+1}$  are in different met zones, a fall time is calculated for each zone

$$T_{f_{sn}}(Z) = \frac{HOB_n - H_{Z-1}}{|v_{sn,2}|} \quad (E.22)$$

$$T_{f_{sn}}(z) = \frac{H_z - H_{z-1}}{|v_{sn,2}|} \quad (E.23)$$

$$T_{f_{sn}}(Z_0) = \frac{H_{Z_0} - HOB_{n+1}}{|v_{sn,2}|} \quad (E.24)$$

where  $Z$  is the met zone of  $HOB_n$ ,  $Z_0$  is the met zone of  $HOB_{n+1}$  and  $z$  is any met zone  $Z < z < Z_0$ .  $H_z$  is the upper height of met zone  $z$ . When  $n$  is the last submunition phase, equation (E.21) becomes zero.

The displacement due to horizontal wind is the sum of contributions from applicable met zones.

$$\vec{d} = \sum_{s=Z_0}^Z T_{f_{sn}}(z) \vec{w}_h(z) \quad (E.25)$$

where  $\vec{d}$  is the displacement vector and  $\vec{w}_h(z)$  the horizontal wind vector of met zone  $z$  (assumed constant throughout the zone).

The displacement due to wind may for certain types of submunition, e.g. illumination canisters, be used to make an offset to the aim point. The offset is

$$\vec{o} = -\vec{d} \min \left( a, \frac{o_{\max}}{|\vec{d}|} \right) \quad (E.26)$$

where:  $\vec{o}$  is the offset vector,  $a \leq 0.5$  is a factor indicating the relation between the displacement and the offset. The offset should not exceed a certain horizontal distance  $o_{max}$ . The minus sign indicates that the offset vector is in the opposite direction as the horizontal wind vector.

For illumination canisters the maximum offset is

$$o_{max} = \sqrt{R_{ill}^2 - HOB^2} \quad (E.27)$$

where:  $R_{ill}$  is the illumination radius (maximum absolute distance between illumination canister and target). HOB is the height at which the illumination canister ignites.

or

c. Displacement

by fitting observed data (for phase 1) to the form

$$\Delta X_{1,1} = a_0 + a_1 v_{carrier,1} = \text{along the horizontal component of } \vec{v} \quad (E.28)$$

$$\Delta X_{1,2} = b_0 + b_1 v_{carrier,2} = \text{along the vertical component of } \vec{v} \quad (E.29)$$

$$\Delta t = c_0 = \text{time displacement} \quad (E.30)$$

or

d. Closed Form Biases

The offset is calculated using closed form rule for certain types of sensor-fuzed munitions

This describes the change of the submunition data from the ejection of the carrier shell to the time where a deceleration device is fully deployed.

The change of the flight conditions from the base separation point after ejection to the point where the deceleration devices (eg. ballutes, parachutes) are fully deployed, are calculated by polynomial functions.

The separation phase is described by the following equations, where the index  $s$  represents the submunition.

For a detailed description of the coefficients used, see chapter List of Symbols.

The change in position  $\Delta \vec{X}_s$  is given as

$$\Delta \vec{X}_s = a_{0s} \begin{bmatrix} \cos \Gamma_0 \cos \psi_0 \\ \sin \Gamma_0 \\ \cos \Gamma_0 \sin \psi_0 \end{bmatrix} \quad (E.31)$$

where  $\Gamma_0$  is the vertical angle of fall and  $\psi_0$  is the horizontal angle of fall at the start of the phase.

The change in velocity is given as

$$\Delta u_s = a_0 + a_1 u_0 + a_2 \Gamma_0 + a_3 u_0 \Gamma_0 \quad (\text{E.32})$$

where  $u_0$  is the velocity and  $\Gamma_0$  is the vertical angle of fall at the start of the phase.

The change in the vertical angle of fall is given as

$$\Delta \Gamma_s = a_0 + a_1 u_0 + a_2 u_0^2 + (b_0 + b_1 u_0 + b_2 u_0^2) \Gamma_0 + (c_0 + c_1 u_0 + c_2 u_0^2) \Gamma_0^2 \quad (\text{E.33})$$

where  $u_0$  is the velocity and  $\Gamma_0$  is the vertical angle of fall at the start of the phase.

The change in spin is given by

$$\Delta p_s = a_0 + a_1 p_0 + a_2 u_0 + a_3 p_0 u_0 \quad (\text{E.34})$$

where  $p_0$  is the spin and  $u_0$  is the velocity at the start of the phase.

The duration of the phase is given by

$$\Delta t_s = a_0 + a_1 u_0 + a_2 u_0^2 \quad (\text{E.35})$$

where  $u_0$  is the velocity at the start of the phase.

## 5. Calculation of Submunition Dispersion

### a. Concentric submunitions

The axis of the shell will not in general be parallel to the shell velocity vector due to a non-zero yaw of repose. This effect may be accounted for as shown in the following dispersion calculation.

The axis of the carrier shell is the vector  $\vec{c}$  and the yaw of repose is defined by the vector  $\vec{\alpha}_e$ . Thus we have

$$\vec{c} = \frac{\vec{v}}{v} \cos \alpha_e + \vec{\alpha}_e \quad (\text{E.36})$$

The ejection velocity of any central bomblet (a bomblet lying concentric with the axis of the shell), we get

$$\vec{u}_{s1} = \vec{u}_{carrier} + \Delta u_{s1} \vec{c} \quad (\text{E.37})$$

where:

$$\Delta u_{s1} = a_0 + a_1 p \quad (\text{E.38})$$

is the ejection speed of the submunition, and  $p$  is the magnitude of projectile's spin at time of fuze function.

b. Eccentric submunitions

For a cross-sectional layer of submunitions, the ejection velocity of eccentric submunitions in the plane normal to the shell axis may be calculated by

$$\vec{u}_{s, normal} = i_D (\vec{p} \times \vec{r}_s) \quad (E.39)$$

where:  $\vec{p}$  is the spin vector at base separation and  $\vec{r}_s$  is the vector from the axis of the carrier shell to the center of gravity of the individual submunition.  $i_D$  is a fitting factor. The configuration is shown in figure E-2.

In order to find the ejection velocity we first have to find the horizontal vector  $\vec{e}$  that is normal to  $\vec{c}$  and the vector  $\vec{d} = \vec{e} \times \vec{c}$ . These are also shown in figure E-2. The vector  $\vec{e}$  can be written as

$$\vec{e} = \frac{\vec{c} \times \vec{y}}{|\vec{c} \times \vec{y}|} \quad (E.40)$$

where  $\vec{y}$  is defined as a unit vector in the positive 2-direction.

If we now define  $\phi$  as the angle between the vector  $\vec{d}$  and the position of the submunition as shown in figure E-2, the position vector of the submunition from the axis is

$$\vec{r}_s = r_s (\vec{e} \sin \phi + \vec{d} \cos \phi) \quad (E.41)$$

In the modified point mass model, we always have  $\vec{p} = p\vec{c}$  and when inserted into (E.39)

$$\vec{u}_{s, normal} = i_D (p\vec{c} \times \vec{r}_s) \quad (E.42)$$

After combining (E.38-39) and adding the velocity for the central bomblets we get

$$\vec{u}_{s1} = \vec{u}_{carrier} + \Delta u_{s1} \vec{c} + i_D p r_s (-\vec{d} \sin \phi + \vec{e} \cos \phi) \quad (E.43)$$

where:

$$\Delta u_{s1} = a_0 + a_1 p \quad (E.44)$$

is the ejection speed of the submunition, and  $p$  is the magnitude of the projectile's spin at the time of fuze function.

In the case when  $|\vec{c} \times \vec{y}| = 0$ , both  $\vec{d}$  and  $\vec{e}$  are in the horizontal plane. In this case, choose  $\vec{d}$  to be the unit vector in the direction of the projection of the initial projectile velocity onto the horizontal plane.

## c. Submunition trajectories

The ballistic trajectory of the submunitions can be determined by the same methods as described in sections 3 and 4.

This pattern is defined by the impact point of four virtual submunitions defined relative to the vertical plane of the trajectory of the carrier:

- (1) the submunition ejected in the vertical plane above the shell trajectory,
- (2) the submunition ejected in the vertical plane below the shell trajectory,
- (3) the submunition whose tangential ejection velocity component is normal to the vertical plane in the negative 3-direction,
- (4) the submunition whose tangential ejection velocity component is normal to the vertical plane in the positive 3-direction.

The initial velocities for these submunitions at the base separation point are respectively

$$\vec{u}_{lower} = \vec{u}_{carrier} + \Delta u_{s1} \vec{c} - i_D pr_s \vec{e} \quad (\phi = 180^\circ) \quad (E.45)$$

$$\vec{u}_{upper} = \vec{u}_{carrier} + \Delta u_{s1} \vec{c} + i_D pr_s \vec{e} \quad (\phi = 0^\circ) \quad (E.46)$$

$$\vec{u}_{right} = \vec{u}_{carrier} + \Delta u_{s1} \vec{c} - i_D pr_s \vec{d} \quad (\phi = 90^\circ) \quad (E.47)$$

$$\vec{u}_{left} = \vec{u}_{carrier} + \Delta u_{s1} \vec{c} + i_D pr_s \vec{d} \quad (\phi = 270^\circ) \quad (E.48)$$

In cases (3) and (4) it is anticipated that the ejection takes place at such a height that the descent is close to vertical at the end of the terminal phase.

## 6. Empty Carrier Trajectory

The trajectory of the empty carrier left after the base separation point can be calculated assuming that the carrier has the same aerodynamic properties as the original shell. The empty carrier will at base separation get a velocity, a mass and a moment of inertia, which are different to the original shell. The calculation of the trajectory could be done according to the Modified Point Mass Model in the main text or the Point Mass Model in Annex F.

The velocity of the empty carrier is found by

$$\vec{u}_{ec} = \vec{u}_{carrier} + \Delta u_{ec} \vec{c} \quad (E.49)$$

The sign of  $\Delta u_{ec}$  is opposite to  $\Delta u_{s1}$ . The mass of the empty carrier  $m_{ec}$ , its moment of inertia  $I_{x_{ec}}$ , and  $\Delta u_{ec}$  must be entered separately.

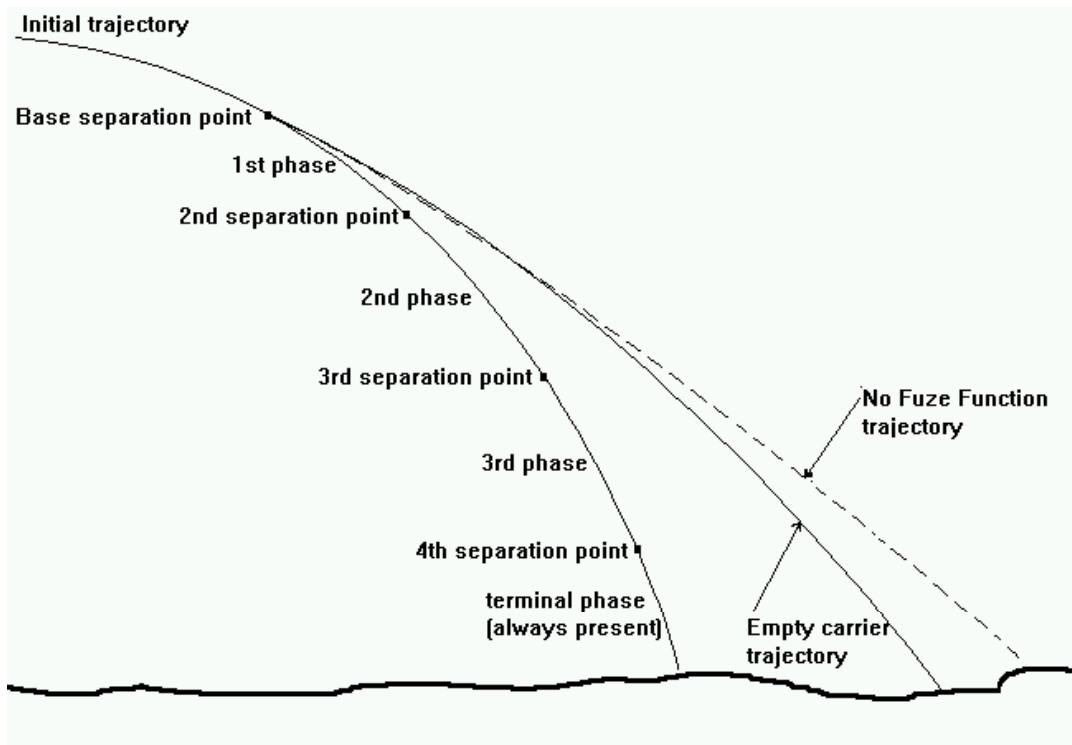


Figure E-1 Definition of concepts for submunition trajectories

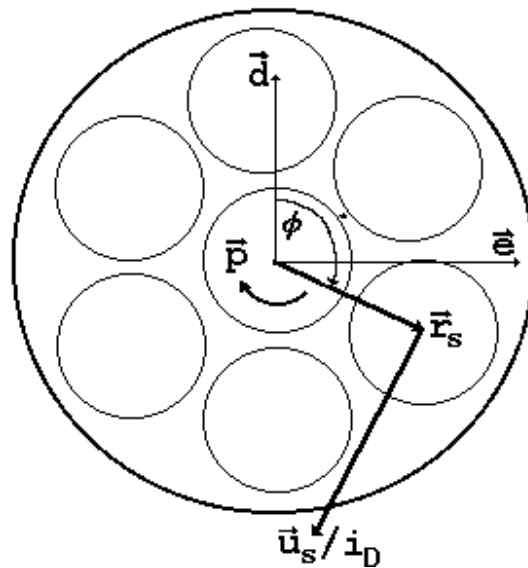


Figure. E-2 Geometry of tangential ejection velocity

7. List of Additional Symbols

| <u>Symbol</u>                   | <u>Definition</u>                                              | <u>Units</u>         |
|---------------------------------|----------------------------------------------------------------|----------------------|
| $Alt_w$                         | Altitude of weapon above sea level                             | <i>m</i>             |
| $A_s$                           | Angle of site                                                  | <i>mil</i>           |
| $a_0, a_1 \dots b_0, b_1 \dots$ | Auxiliary coefficients                                         | <i>none</i>          |
| $\vec{c}$                       | Axis of carrier shell                                          | <i>none</i>          |
| $C_D$                           | Drag coefficient                                               | <i>none</i>          |
| $C_{spin}$                      | Spin damping coefficient                                       | <i>none</i>          |
| $\vec{d}$                       | Displacement vector due to wind                                | <i>m</i>             |
| $HOB$                           | Height of burst                                                | <i>m</i>             |
| $H_z$                           | Upper height of met zone <i>z</i>                              | <i>m</i>             |
| $i_D$                           | Fitting factor for dispersion of submunition                   | <i>none</i>          |
| $m_s$                           | Mass of submunition unit                                       | <i>kg</i>            |
| $\vec{o}$                       | Offset vector                                                  | <i>m</i>             |
| $QE$                            | Quadrant elevation                                             | <i>mil</i>           |
| $r_s$                           | Distance from carrier axis to center of gravity of submunition | <i>m</i>             |
| $S$                             | Reference cross sectional area                                 | <i>m<sup>2</sup></i> |
| $t$                             | Simulation time since start of phase                           | <i>s</i>             |
| $T_f$                           | Fall time of submunition (duration of phase)                   | <i>s</i>             |
| $T_s$                           | Time since start of phase                                      | <i>s</i>             |
| $\vec{w}_h$                     | Horizontal wind velocity                                       | <i>m/s</i>           |
| $\vec{x}_n$                     | Position at <i>n</i> th separation point                       | <i>m</i>             |

|            |                                                                                                |                         |
|------------|------------------------------------------------------------------------------------------------|-------------------------|
| $z$        | Met zone index                                                                                 | <i>none</i>             |
| $\Delta u$ | Axial ejection velocity of submunition or velocity change of empty carrier due to ejection     | <i>m/s</i>              |
| $\Gamma_0$ | Vertical angle of fall at the start of the phase                                               | <i>rad</i>              |
| $\phi$     | Positional angle of submunition at ejection (clockwise from positive vertical, see figure E-2) | <i>rad</i>              |
| $\psi_0$   | Horizontal angle of fall at the start of the phase                                             | <i>rad</i>              |
| $\rho$     | Density of air                                                                                 | <i>kg/m<sup>3</sup></i> |

Subscripts

(multiple subscripts occur)

*carrier*

property of carrier shell at base separation

*ec*

property of empty carrier shell

*n* (or a *digit*)

property in terminal phase *n* (or phase number as given by *digit*)

*s*

property of submunition

1, 2, 3

(when following a comma) component in the 1-2-3 system

Symbols without vector indicator symbolize the scalar value of the vector



### POINT MASS TRAJECTORY MODEL

1. The following equations constitute a mathematical model representing the flight of a projectile. The modeling is accomplished by simplifying the modified point mass equations given in Section I of this document to those representing a point having mass and spin only.
2. Newton's law of motion for a point mass is:

$$\vec{F} = m\dot{\vec{u}} = \overrightarrow{DF} + m\vec{g} + m\vec{\Lambda} \quad (\text{F.1})$$

where:

acceleration due to drag force is:

when  $i$  is used as a fitting factor,

$$\frac{\overrightarrow{DF}}{m} = - \left( \frac{\pi \rho i d^2}{8 m} \right) C_D v \vec{v} \quad (\text{F.2})$$

or when  $C$  is used as a fitting factor,

$$\frac{\overrightarrow{DF}}{m} = - \left( \frac{\pi \rho m_r}{8 C m} \right) C_D v \vec{v} \quad (\text{F.3})$$

or when  $f_D$  is used as a fitting factor,

$$\frac{\overrightarrow{DF}}{m} = - \left( \frac{\pi \rho d^2}{8 m} \right) (f_D C_D) v \vec{v} \quad (\text{F.4})$$

acceleration due to gravity is:

$$\vec{g} = -g_0 \left( \frac{R^2}{r^3} \right) \vec{r} = -g_0 \begin{bmatrix} \frac{X_1}{R} \\ 1 - \frac{2X_2}{R} \\ \frac{X_3}{R} \end{bmatrix} \quad (\text{F.5})$$

$$\text{where: } g_0 = 9.80665 [ 1 - .0026 \cos (2 \text{ lat}) ] \quad (\text{F.6})$$

$$\vec{r} = \vec{X} - \vec{R}, \quad (\text{F.7})$$

$$\vec{R} = \begin{bmatrix} 0 \\ -R \\ 0 \end{bmatrix} \quad (\text{F.8})$$

and  $R = 6.356766 \times 10^6$  meters

acceleration due to the effect of the Coriolis force is:

$$\vec{A} = -2(\vec{\omega} \times \vec{u}) \quad (\text{F.9})$$

where:

$$\vec{\omega} = \begin{bmatrix} \Omega \cos(lat) \cos(AZ) \\ \Omega \sin(lat) \\ -\Omega \cos(lat) \sin(AZ) \end{bmatrix} \quad (\text{F.10})$$

and  $\Omega = 7.292115 \times 10^{-5} \text{ rad/s}$

NOTE: For southern hemisphere, *lat* is negative.

3. The magnitude of spin acceleration is given by:

$$\dot{p} = \frac{\pi \rho d^4 p v C_{spin}}{8 I_x} \quad (\text{F.11})$$

where:

$$p = p_0 + \int_0^t \dot{p} dt \quad (\text{F.12})$$

is the magnitude of spin at time =  $t$  and

$$p_0 = \frac{2\pi u_0}{t_c d} \quad (\text{F.13})$$

is the magnitude of the initial spin of the projectile at the muzzle.

4. The velocity of the projectile with respect to the ground fixed axes at time = t is:

$$\vec{u} = \vec{u}_0 + \int_0^t \dot{\vec{u}} dt \quad (\text{F.14})$$

where:

$$\vec{u}_0 = \begin{bmatrix} u_0 \cos(QE) \cos(\Delta AZ) \\ u_0 \sin(QE) \\ u_0 \cos(QE) \sin(\Delta AZ) \end{bmatrix} \quad (\text{F.15})$$

5. The velocity of the projectile with respect to the air is given by:

$$\vec{v} = \vec{u} - \vec{w} \quad (\text{F.16})$$

and Mach number by:

$$M = v / a \quad (\text{F.17})$$

where  $a$  is the speed of sound as defined at paragraph 21 on Page 14.

6. The position of the projectile with respect to the ground axes is given by:

$$\vec{X} = \vec{X}_0 + \int_0^t \vec{u} dt \quad (\text{F.18})$$

where:

$$\vec{X}_0 = \begin{bmatrix} l_w \cos(QE) \cos(\Delta AZ) \\ X_{2w} + l_w \sin(QE) \\ l_w \cos(QE) \sin(\Delta AZ) \end{bmatrix} \quad (\text{F.19})$$

is the location of the muzzle, i.e., initial position of the center of mass of the projectile.

7. The position of the projectile with respect to the spherical earth's surface is given by the approximation:

$$\vec{E} = \begin{bmatrix} X_1 \\ X_2 + \frac{(X_1^2 + X_3^2)}{2R} \\ X_3 \end{bmatrix} \quad (\text{F.20})$$

8. For some applications, especially where the flight path of the projectile is short, the model described above may be simplified further:

the effects of height above the earth's surface on gravity may be ignored, in which case

$$\vec{g} = -g_0 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (\text{F.21})$$

the effects of spin of the earth may be ignored, in which case

$$\Omega \equiv 0 \quad (\text{F.22})$$

the effects of projectile spin may be ignored, in which case

$$p_0 = \dot{p} \equiv 0 \quad (\text{F.23})$$

the earth's surface may be considered a flat plane, in which case

$$\vec{E} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \quad (\text{F.24})$$

9. The List of Symbols found at Sections II and III of the main document contain all the symbols required for this annex with one exception, namely,  $C_D$  is defined as the total drag force coefficient (dimensionless).
10. The List of Data Requirements related to this annex are as follows:

| <b><i>Physical Data</i></b>                                          | <b><i>Symbol</i></b> |
|----------------------------------------------------------------------|----------------------|
| Initial speed of projectile with respect to ground (muzzle velocity) | $u_0$                |
| Rifling twist at muzzle                                              | $t_c$                |
| Mass of fuzed projectile                                             | $m$                  |
| Mass of fuzed reference projectile                                   | $m_r$                |
| Reference diameter                                                   | $d$                  |
| Initial axial moment of inertia (optional)                           | $I_{x_0}$            |

| <b><i>Aerodynamic Data</i></b>             | <b><i>Symbol</i></b> |
|--------------------------------------------|----------------------|
| Total drag force coefficient               | $C_D$                |
| Spin damping moment coefficient (optional) | $C_{spin}$           |

## a. Fitting Factors for Range Estimation

Since the point mass equations do not explicitly model the effects of yaw of repose, the fitting factors for the range estimation must be used to account for this fact. As a result, the forms will vary from those forms used with the MPM equations. Following are some typical forms used in fire control systems.

## Quadrant Elevation Fitting

Ballistic coefficient (C):

$$C = a_0 + a_1(QE) + a_2(QE)^2 + a_3(QE)^3 \quad (F.25)$$

Form factor (i):

$$i = b_0 + b_1(QE) + b_2(QE)^2 + b_3(QE)^3 \quad (F.26a)$$

or

$$i = b_0 + b_1(QE - A_s) + b_2(QE - A_s)^2 + b_3(QE - A_s)^3 \quad (F.26b)$$

The equations are applicable over the QE intervals bounded by  $QE_{j-1}$  to and including  $QE_j$ . The functions must be continuous and smooth at each breakpoint. The minimum and maximum values of QE are determined by the weapon and ammunition characteristics but, typically, range from zero mils to the maximum trail angle. The degree of the polynomial is dependent upon the desired quality of the range estimation.

## Mach Number Fitting

Drag factor ( $f_D$ ) can be fitted as a function of Mach number.

## b. Drift Correction

Since the point mass equations do not include the drift effects associated with spinning projectiles, the drift (which is perpendicular to the vertical plane containing the line of fire) is generally represented by a function based on the observed fall of shot results. The following are typical expressions for drift expressed in mil:

$$Drift = \tan(QE) f(QE) \quad (F.27)$$

where:

$$f(QE) = a_0 + a_1(QE) + a_2(QE)^2 + a_3(QE)^3 \quad (F.28)$$

or

$$Drift = b_0 \tan(QE) + b_1 \tan^2(QE) \quad (F.29)$$

or

$$Drift = c_0 QE / (QE + c_1) \quad (F.30)$$

or

$$Drift = \arctan ((d_0 + d_1 T + d_2 T^2 + d_3 T^3)/x_l) \quad (F.31)$$

where:  $x_l$  is the range (m) to impact along the  $\bar{l}$  axis.

c. Time of Flight Correction

In order to account for the difference between the actual time of flight (T) and the time of flight predicted by the point mass model the following forms can be used:

$$T = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \quad (F.32)$$

Or, if approximating  $\Delta t$  as a function of  $t$ ,

$$T = t + \Delta t \quad (F.33)$$

and 
$$\Delta t = b_0 + b_1 t + b_2 t^2 + b_3 t^3 \quad (F.34)$$

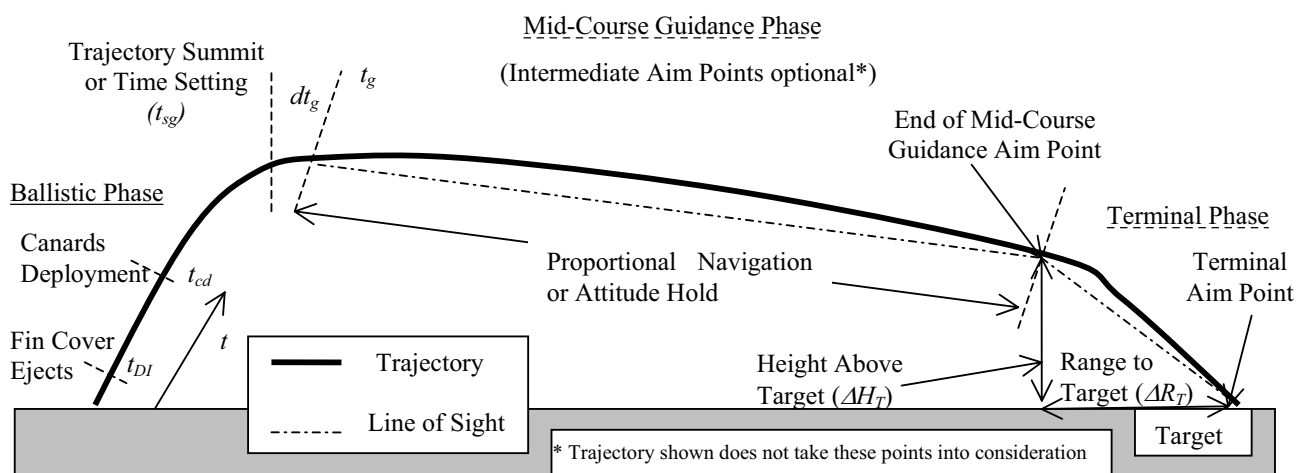
Where the equation is applicable over the time (t) intervals bounded by  $t_{j-1}$  to and including  $t_j$ . The functions must be continuous and smooth at each breakpoint. The minimum and maximum values of T are determined by the weapon and ammunition characteristics but, typically, range from zero seconds to the maximum time of flight. The degree of the polynomial is dependent upon the desired quality of the time estimation.

## ADDITIONAL TERMS FOR GUIDED MUNITIONS

### 1. Introduction

This annex presents a method of extending the use of a modified point mass trajectory model for mid-course guidance munitions utilizing Global Positioning System (GPS) and Inertial Navigation System (INS), which incorporates an estimate of the vertical and horizontal angles of yaw to simulate the flight of the guided munitions. The equations in this annex are to be used during the guided and terminal phases of the trajectory, where the munition departs from a ballistic trajectory ( $t \geq t_g$ ).

The figure below shows the phases of a trajectory for guided munitions. This trajectory shows one Aim Point after the start of guidance in the Mid-Course Guidance Phase with a line of sight drawn between the two points. If more than one Aim Point is desired in this phase, each Aim Point will have a line of sight vector drawn from the previous Aim Point to the following Aim Point. The Proportional Navigation methods described in this annex hold for each Aim Point in the Mid Course Guidance Phase.



**Figure G-1. Phases of Guided Munition Trajectory**

If the canards are deployed at time  $t_{cd}$  during the ballistic phase, the form factor and zero-yaw drag force coefficient are  $i_{b_{cd}}$  and  $C_{D0g}$  respectively.

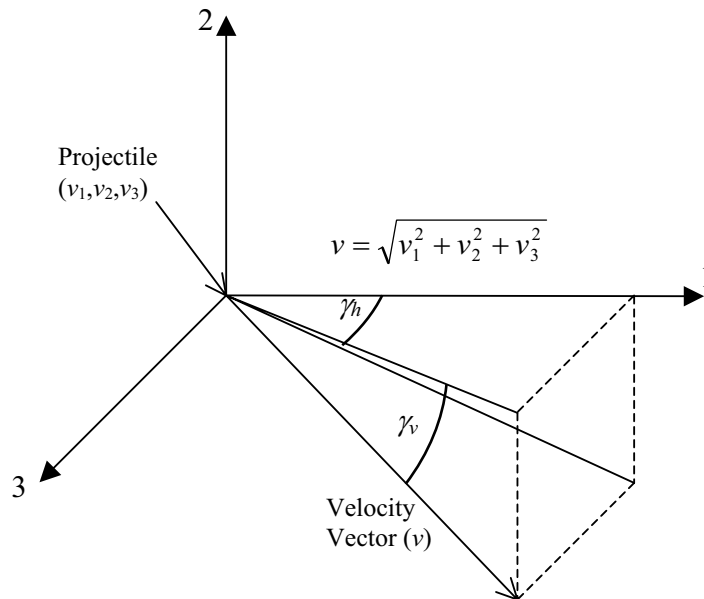
## 2. Pointing of Munition

The pointing of the munition is assumed to be the same as the pointing of the velocity vector. The projection of the velocity vector on the vertical ( $\gamma_v$ ) and horizontal ( $\gamma_h$ ) planes is given by:

$$\gamma_v = \arcsin\left(\frac{v_2}{v}\right) \quad (\text{G.1})$$

$$\gamma_h = \arctan\left(\frac{v_3}{v_1}\right) \quad (\text{G.2})$$

Then define a unit vector along the longitudinal axis of the munition including an estimate of the angles of yaw to simulate the guidance technologies during the guidance phase.



**Figure G-2.** Angles for Pointing of Munition



### 3. Proportional Navigation Guidance to an Aim Point

The Aim Point is defined by a height above the target ( $\Delta H_T$ ) and a distance in range from the target ( $\Delta R_T$ ).

where:

$$\Delta R_T = a_0 + a_1(\lambda_{v_{ig}}) + a_2(\lambda_{v_{ig}})^2 \quad (G.3)$$

$$\Delta H_T = a_0 + a_1(\lambda_{v_{ig}}) + a_2(\lambda_{v_{ig}})^2 \quad (G.4)$$

The unit vector along the longitudinal axis of the munition to proportionally navigate to an aim point beginning at the initiation of guidance ( $t_g$ ) is given by:

$$\vec{x} = \begin{bmatrix} \cos(\gamma_v + f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}}) \cos \left[ \gamma_h + \left( \frac{f_{\alpha_h} \alpha_{h_{pn}}}{\cos \gamma_v} \right) \right] \\ \sin(\gamma_v + f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}}) \\ \cos(\gamma_v + f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}}) \sin \left[ \gamma_h + \left( \frac{f_{\alpha_h} \alpha_{h_{pn}}}{\cos \gamma_v} \right) \right] \end{bmatrix} \quad (G.5)$$

where:

$$\alpha_{v_{pn}} = - \frac{8 m N_v \dot{\lambda}_v}{\pi \rho d^2 \left( C_{L_{\alpha_g}} + C_{L_{\alpha_g^3}} \alpha_g^2 \right) v} \quad (G.6)$$

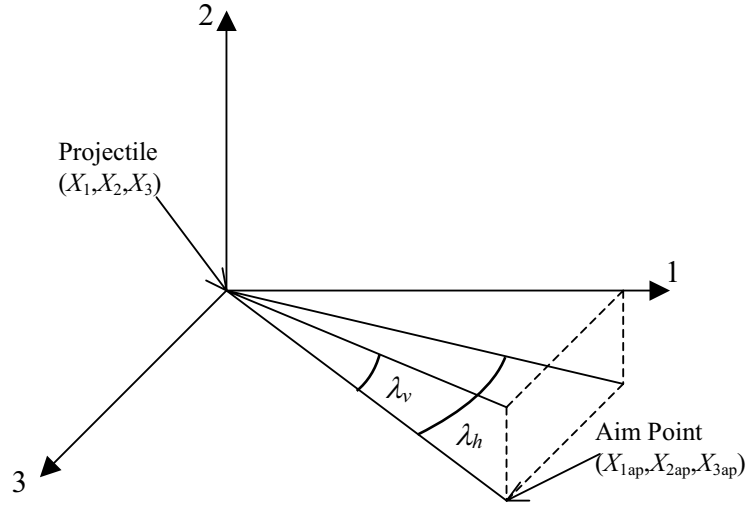
$$\alpha_{h_{pn}} = - \frac{8 m N_h \dot{\lambda}_h}{\pi \rho d^2 \left( C_{L_{\alpha_g}} + C_{L_{\alpha_g^3}} \alpha_g^2 \right) v} \quad (G.7)$$

$N_v$  and  $N_h$  are the vertical and horizontal plane navigation ratios, respectively. They are linear polynomials of  $(t - t_g)$  in the form  $a_0 + a_1(t - t_g)$ .

The pointing angle or angles of line-of-sight vector (from munition to aim point) in inertial space are obtained from the known aim point position and the munition position (for all aim points in the guidance phase) from GPS by:

$$\lambda_v = \arctan \left( \frac{X_{2_{ap}} - X_2}{\sqrt{(X_{1_{ap}} - X_1)^2 + (X_{3_{ap}} - X_3)^2}} \right) \quad (G.8)$$

$$\lambda_h = \arctan \left( \frac{X_{3_{ap}} - X_3}{\sqrt{(X_{1_{ap}} - X_1)^2 + (X_{2_{ap}} - X_2)^2}} \right) \quad (G.9)$$



**Figure G-3.** Pointing Angles (Line of Sight Vector)

Then the rate of change of the pointing angles to the aim point are given by:

$$\dot{\lambda}_v = \frac{\left[ (X_{1_{ap}} - X_1)^2 + (X_{3_{ap}} - X_3)^2 \right] \dot{X}_2 - (X_{2_{ap}} - X_2) \left[ (X_{1_{ap}} - X_1) \dot{X}_1 + (X_{3_{ap}} - X_3) \dot{X}_3 \right]}{\left[ (X_{1_{ap}} - X_1)^2 + (X_{2_{ap}} - X_2)^2 + (X_{3_{ap}} - X_3)^2 \right] \sqrt{(X_{1_{ap}} - X_1)^2 + (X_{3_{ap}} - X_3)^2}} \quad (\text{G.10})$$

$$\dot{\lambda}_h = \frac{\left[ (X_{1_{ap}} - X_1)^2 + (X_{2_{ap}} - X_2)^2 \right] \dot{X}_3 - (X_{3_{ap}} - X_3) \left[ (X_{1_{ap}} - X_1) \dot{X}_1 + (X_{2_{ap}} - X_2) \dot{X}_2 \right]}{\left[ (X_{1_{ap}} - X_1)^2 + (X_{2_{ap}} - X_2)^2 + (X_{3_{ap}} - X_3)^2 \right] \sqrt{(X_{1_{ap}} - X_1)^2 + (X_{2_{ap}} - X_2)^2}} \quad (\text{G.11})$$

and gravity bias ( $\alpha_{v_{gb}}$ ), which is determined as a function of time, is given by:

$$\alpha_{v_{gb}} = - \frac{8 g_2 m (\cos \gamma_v)}{\pi \rho d^2 \left( C_{L_{\alpha_g}} + C_{L_{\alpha_g^3}} \alpha_g^2 \right) v^2} \quad (\text{G.12})$$

NOTE:  $g_2$  represents the vertical component of the gravity vector.

To compensate for the approximations in the equations of motion, the fitting factors  $f_{\alpha_v}$ ,  $f_{\alpha_h}$ , and  $f_{gb}$  are applied in order to match the computed and observed results. The factor on gravity bias ( $f_{gb}$ ) is reduced linearly starting at a distance from the aim point to 0 at a closer distance to the aim point. Fitting factors are defined in Table G-1.

The total angle of yaw during the guidance phase,  $\vec{\alpha}_g$ , is given by:

$$\vec{\alpha}_g = \vec{x} - (\vec{x} \cdot \vec{v}) \left( \frac{\vec{v}}{v^2} \right) \quad (G.13)$$

#### 4. Maximum Angles of Yaw

The total angle of yaw combining the horizontal and vertical directions will not be allowed to exceed certain constraints. A vertical upward angle of attack  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}})$  will increase the range by increasing lift and reduce the range by increasing the drag. These opposing effects are balanced to avoid a decrease in maximum range by defining a vertical upward angle of attack limit  $\alpha_{vu_{max}}$  that yields the maximum lift-to-drag ratio; and  $(f_{\alpha_h} \alpha_{h_{pn}})$  must vanish as  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}})$  approaches  $\alpha_{vu_{max}}$  or the total drag will be increased beyond the maximum lift-to-drag ratio.

Therefore, for  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}}) > 0$ : the limitation on the vertical upward angle of yaw  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}})$  is  $\alpha_{vu_{max}}$ . The limitation on  $(f_{\alpha_h} \alpha_{h_{pn}})$  will be dependent on the magnitude of  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}})$ . If  $\alpha_{vu_{max}}$  is the limit on  $(f_{\alpha_h} \alpha_{h_{pn}})$  when  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}}) = 0$ , then when  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}}) > 0$ ,  $\alpha_{vu_{max}}$  must be reduced to prevent  $(f_{\alpha_h} \alpha_{h_{pn}})$  from adding drag that would compromise the maximum lift-to-drag criteria.

The horizontal angle of attack is limited as follows:

$$\left| (f_{\alpha_h} \alpha_{h_{pn}})_{\max} \right| = \alpha_{h_{\max}} \sqrt{1 - \frac{(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}})^2}{\alpha_{vu_{\max}}^2}} \quad (G.14)$$

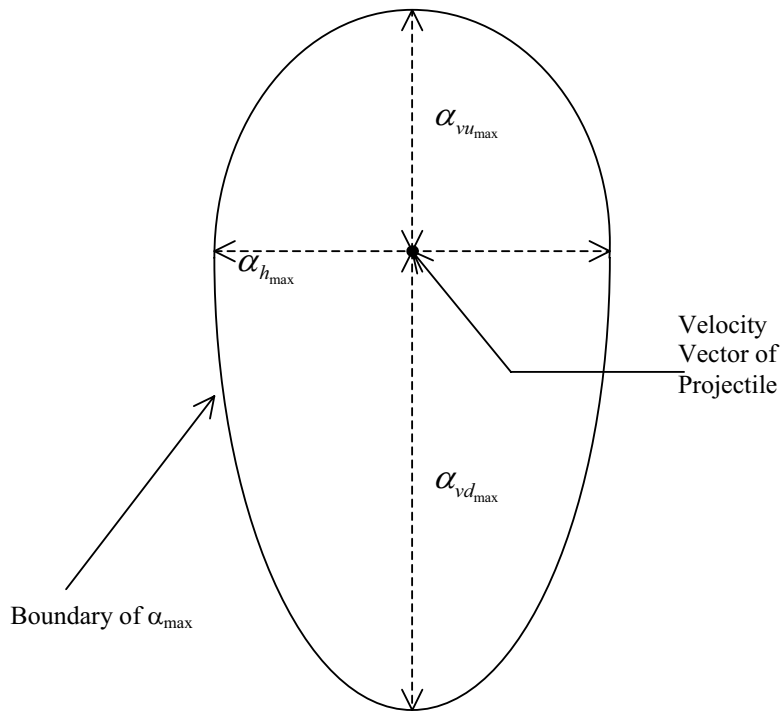
For  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}}) \leq 0$ : the maximum lift-to-drag ratio no longer applies, since the downward lift and drag no longer oppose one another, so the primary limiter can be  $(f_{\alpha_h} \alpha_{h_{pn}})$ .

Therefore, for  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}}) \leq 0$ : the limitation on  $(f_{\alpha_h} \alpha_{h_{pn}})$  will be  $\alpha_{h_{\max}}$  and the limitation on  $(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}})$  will be dependent on the magnitude of  $(f_{\alpha_h} \alpha_{h_{pn}})$  and the maximum vertical downward angle of attack  $\alpha_{vd_{\max}}$ .

The vertical downward angle of attack is limited as follows:

$$(f_{\alpha_v} \alpha_{v_{pn}} + f_{gb} \alpha_{v_{gb}})_{\max} = -\alpha_{vd_{\max}} \sqrt{1 - \frac{(f_{\alpha_h} \alpha_{h_{pn}})^2}{\alpha_{h_{\max}}^2}} \quad (G.15)$$

The maximum vertical upward ( $\alpha_{vu_{\max}}$ ), vertical downward ( $\alpha_{vd_{\max}}$ ), and horizontal ( $\alpha_{h_{\max}}$ ) angles of attack are quartic polynomials of Mach number.



**Figure G-4.** Maximum Angles of Yaw

## 5. Maintain Munition Pointing (Attitude Hold)

To maintain munition pointing, the unit vector along the longitudinal axis of the munition that existed at the initiation of glide ( $t_g$ ), that is a time interval ( $dt_g$ ) after a time setting ( $t_{sg}$ ), is given by:

$$\vec{x} = \begin{Bmatrix} \cos \gamma_{v_{t_g}} & \cos \gamma_{h_{t_g}} \\ \sin \gamma_{v_{t_g}} & \\ \cos \gamma_{v_{t_g}} & \sin \gamma_{h_{t_g}} \end{Bmatrix} \quad (G.16)$$

The unit vector remains fixed as above as long as  $\alpha_g$ , the total angle of yaw during the guidance phase, is  $\leq \alpha_{vu_{max}}$ . When  $\alpha_g > \alpha_{vu_{max}}$  due to the path of the trajectory falling away from the pointing vector of the munition, the attitude must be allowed to drift downward to prevent  $\alpha_g$  from becoming too large and causing the munition to stall.

During guidance, when  $\alpha_g > \alpha_{vu_{max}}$  the unit vector is:

$$\vec{x} = \begin{Bmatrix} \cos(\gamma_{v_{t_g}} + \alpha_{vu_{max}}) \cos \gamma_{h_{t_g}} \\ \sin(\gamma_{v_{t_g}} + \alpha_{vu_{max}}) \\ \cos(\gamma_{v_{t_g}} + \alpha_{vu_{max}}) \sin \gamma_{h_{t_g}} \end{Bmatrix} \quad (G.17)$$

## 6. Fitting Factors

**Table G-1. Fitting Factors**

| Fitting Factor Data Requirement | Symbol         |
|---------------------------------|----------------|
| Vertical angle of yaw           | $f_{\alpha_v}$ |
| Gravity Bias angle of yaw       | $f_{gb}$       |
| Horizontal angle of yaw         | $f_{\alpha_h}$ |

## 7. Forms of Fire Control Input Data

### a. Functions of Horizontal Inertial Line-of-sight Angle:

Fitting factor for Horizontal angle of yaw ( $f_{\alpha_h}$ ) is given for each charge as follows:

$$f_{\alpha_h} = a_0 + a_1(\lambda_h) + a_2(\lambda_h)^2 \quad (G.18)$$

where:  $\lambda_h$  is evaluated at  $t_g$

### b. Functions of Vertical Inertial Line-of-sight Angle:

Fitting factors for Vertical angle of yaw ( $f_{\alpha_v}$ ) and Gravity Bias angle of yaw ( $f_{gb}$ ) are given for each charge as follows:

$$f_{\alpha_v} = a_0 + a_1(\lambda_v) + a_2(\lambda_v)^2 \quad (G.19)$$

$$f_{gb} = a_0 + a_1(\lambda_v) + a_2(\lambda_v)^2 \quad (G.20)$$

where:  $\lambda_v$  is evaluated at  $t_g$

## 8. List of Additional Symbols

| <u>Symbol</u>     | <u>Definition</u>                                                       | <u>Units</u>             |
|-------------------|-------------------------------------------------------------------------|--------------------------|
| $C_{D0g}$         | Zero yaw drag force coefficient with canards deployed                   | <i>none</i>              |
| $C_{L\alpha_g}$   | Lift force coefficient during guidance phase                            | <i>1/rad</i>             |
| $C_{L\alpha_g^3}$ | Cubic lift force coefficient during guidance phase                      | <i>1/rad<sup>3</sup></i> |
| $dt_g$            | Time between the end of Ballistic Phase and start of guidance ( $t_g$ ) | <i>s</i>                 |
| $f_{\alpha_v}$    | Fitting factor on vertical angle of yaw                                 | <i>none</i>              |
| $f_{\alpha_h}$    | Fitting factor on horizontal angle of yaw                               | <i>none</i>              |

|              |                                                                          |             |
|--------------|--------------------------------------------------------------------------|-------------|
| $f_{gb}$     | Fitting factor on gravity bias                                           | <i>none</i> |
| $i_{bcd}$    | Form factor during ballistic phase with canards deployed                 | <i>none</i> |
| $N_v$        | Vertical plane navigation ratio                                          | <i>none</i> |
| $N_h$        | Horizontal plane navigation ratio                                        | <i>none</i> |
| $t_{cd}$     | Time-of-canards deployment                                               | <i>s</i>    |
| $t_g$        | Time-of-guidance or glide                                                | <i>s</i>    |
| $t_{sg}$     | Time-setting for-guidance or glide                                       | <i>s</i>    |
| $v_1$        | Velocity along 1-axis                                                    | <i>m/s</i>  |
| $v_2$        | Velocity along 2-axis                                                    | <i>m/s</i>  |
| $v_3$        | Velocity along 3-axis                                                    | <i>m/s</i>  |
| $X_1$        | Position of projectile along 1-axis                                      | <i>m</i>    |
| $X_2$        | Position of projectile along 2-axis                                      | <i>m</i>    |
| $X_3$        | Position of projectile along 3-axis                                      | <i>m</i>    |
| $X_{1_{ap}}$ | Position of aim point in the ground-fixed coordinate system along 1-axis | <i>m</i>    |
| $X_{2_{ap}}$ | Position of aim point in the ground-fixed coordinate system along 2-axis | <i>m</i>    |
| $X_{3_{ap}}$ | Position of aim point in the ground-fixed coordinate system along 3-axis | <i>m</i>    |
| $\alpha_g$   | Angle of yaw during guidance                                             | <i>rad</i>  |

|                      |                                                                                         |            |
|----------------------|-----------------------------------------------------------------------------------------|------------|
| $\tilde{\alpha}_g$   | Total angle of yaw during guidance phase                                                | <i>rad</i> |
| $\alpha_{h_{\max}}$  | Maximum horizontal angle of attack                                                      | <i>rad</i> |
| $\alpha_{h_{pn}}$    | Proportional navigation horizontal angle of yaw                                         | <i>rad</i> |
| $\alpha_{\max}$      | Maximum angle of attack                                                                 | <i>rad</i> |
| $\alpha_{vd_{\max}}$ | Maximum vertical downward angle of attack                                               | <i>rad</i> |
| $\alpha_{v_{gb}}$    | Gravity bias vertical angle of yaw                                                      | <i>rad</i> |
| $\alpha_{v_{pn}}$    | Proportional navigation vertical angle of yaw                                           | <i>rad</i> |
| $\alpha_{vu_{\max}}$ | Maximum vertical upward angle of attack                                                 | <i>rad</i> |
| $\Delta H_T$         | Difference in height between the End of<br>Mid-Course Guidance Aim Point and the target | <i>m</i>   |
| $\Delta R_T$         | Difference in range between the End of<br>Mid-Course Guidance Aim Point and the target  | <i>m</i>   |
| $\gamma_h$           | Angle of projection on the horizontal plane                                             | <i>rad</i> |
| $\gamma_v$           | Angle of projection on the vertical plane                                               | <i>rad</i> |
| $\lambda_h$          | Horizontal inertial line-of-sight angle                                                 | <i>rad</i> |
| $\lambda_v$          | Vertical inertial line-of-sight angle                                                   | <i>rad</i> |



## **GLOSSARY OF TERMS**

### **1. Reference Frame and Axes**

All vectors have as a frame of reference a right-handed, orthonormal, ground-fixed, Cartesian coordinate system ( $\vec{1}$ ,  $\vec{2}$ , and  $\vec{3}$ ) where:

- a. the origin is the point where the local vertical line, perpendicular to the surface of the geoid, through the weapon trunnion, intersects the geoid.
- b. the  $\vec{1}$  axis is the intersection of the vertical plane of fire and the horizontal plane and pointing along the gun to target line.
- c. the  $\vec{2}$  axis is parallel to the gravity vector,  $\vec{g}$ , and opposite in direction.
- d. the  $\vec{3}$  axis completes the right-handed coordinate system.

The geoid is the surface within or around the earth that is everywhere normal to the direction of gravity and coincides with mean sea level.

### **2. Geophysical Approximations**

The following geophysical approximations are used in this document.

- a. The geoid is locally approximated by a sphere with a radius (R) of 6,356,766 metres.
- b. The gravitational acceleration,  $\vec{g}$ , has the scalar magnitude

$$g_0 = 9.80665 [1 - .0026 \cos (2 \text{ lat})]$$

at the surface of the above-mentioned sphere and is inversely proportional to the square of the distance from the center of the sphere. The term “*lat*” is the latitude of the origin of the coordinate system.

### **3. Coriolis**

The origin of the coordinate system is fixed to the rotating earth, therefore, a Coriolis acceleration term,  $\vec{A}$ , is included.

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4. Yaw of Repose

The Yaw of Repose is defined by an approximation to the particular solution of the classical linearized equations of yawing motion for a dynamically stable projectile possessing at least trigonal symmetry.

NOTE: Based on comparisons of test firings and modified point mass simulations, it has been determined that quadrant elevations corresponding to a yaw of repose of 0.6 radians provide a good approximation of a weapon system's practical maximum quadrant elevation. Quadrant elevations exceeding this limit can lead to erratic flight behavior reflected by increased dispersion in the deflection plane. Details can be found in the reference, Collings, W.Z. and Lieske, R.F., "A study of Artillery Shell Drift at High Angle of Fire Using Solar Aspect Sensors," BRL-MR-2244, U.S. Army Ballistic Research Laboratory, Aberdeen Proving Ground, Maryland, November 1972.

5. Windage Jump

Windage Jump corrects for the effect of a transient yaw caused by a wind shear. This gives the "Modified Point Mass Model" the ability to perform the correct second order response to wind, range effect due to a cross wind, and the deflection effect due to a range wind.

6. Fitting

To compensate for the approximations in the "Modified Point Mass Model" and the aerodynamic data, certain fitting factors are applied in order to create correspondence between the computed and the observed values of range, deflection and the time of flight.

Two fitting systems are used, quadrant elevation or Mach number as shown below:

**Table H-1.** Fitting Data

| Fitting to                              | Fitting Data as a Function of                        |                                                |
|-----------------------------------------|------------------------------------------------------|------------------------------------------------|
|                                         | Quadrant Elevation<br>[One function for each charge] | Mach Number<br>[Same function for all charges] |
| First Order<br>Range, all<br>elevations | *Form factor: $i$<br>(Drag factor: $f_D = 1$ )       | Drag factor: $f_D$<br>(Form factor: $i = 1$ )  |
| Drift                                   | Lift factor: $f_L$                                   | Lift factor: $f_L$                             |
| Second<br>Order<br>Range, high<br>angle | Yaw drag factor: $Q_D$                               | Yaw drag factor: $Q_D$                         |
| Vertex height<br>and time of<br>flight  | Magnus force factor: $Q_M$                           | Magnus force factor: $Q_M$                     |

\*In many countries a ballistic coefficient,  $C = m_r / i d^2$ , is in use as a fitting factor instead of the nondimensional form factor  $i$ .

**SELECTED BIBLIOGRAPHY**

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| A note on recent progress in the design<br>and analysis of Ballistic Trials of<br>Artillery Shells. P. Fitch, RARDE<br>Branch memo 31/1980                | Short resume of first generation<br>analysis method. Contains references<br>to relevant analysis work on the<br>subject.          |

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| <u>Title</u>                                                                                                                                                                                                | <u>Topic</u>                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| A Study of Design and Analysis Methods For Range and Accuracy Trials. SCICON - Jun 1978                                                                                                                     | Review of work leading to current UK techniques for R&A analysis. Contains references to relevant SCICON work in the area. |
| Adoption of a Standard Canon Artillery Firing Table Format. NATO STANAG 4119                                                                                                                                | Contains description of Firing Table content indicating which parametric variations are of interest.                       |
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| <u>Title</u>                                                                                                                                                                                                                                                                                                                                                                                                     | <u>Topic</u>                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base-Bleed Systems for Gun Projectiles,<br>Nils-Erik Gunners, K. Andersson and<br>R. Hellgren, National Defense Research<br>Institute (FOA) Tumba, Sweden, Chapter<br>16, Volume 109, 1988, Progress in<br>Astronautics and Aeronautics, Gun<br>Propulsion Technology, Published by the<br>American Institute of Aeronautics and<br>Astronautics, Inc., 370 L'Efant<br>Promenade S.W., Washington, D.C.<br>20024 | Fundamental principles of base flow<br>techniques and comparisons with<br>Swedish munitions                                                                              |
| Modified Point Mass Trajectory<br>Simulation for Base-Burn Projectiles.<br>R.F. Lieske and J.E. Danberg,<br>BRL Report No. 3321, U.S. Army<br>Ballistic Research Laboratory, Aberdeen<br>Proving Ground, MD, March 1992 (AD<br>A248292)                                                                                                                                                                          | Extension of the trajectory simulation<br>to include base-burn projectiles for use<br>in fire control systems. Comparisons<br>with observed performance are<br>included. |
| A Physically More Direct and<br>Mathematically More Simple Approach<br>To the MPM Trajectory Model.<br>Prof. In. E. Celens – Head Dept Ballistics<br>and Armament Royal Military Academy<br>- Brussels – 10 <sup>th</sup> International Symposium<br>on Ballistics – San Diego, California,<br>1987                                                                                                              | Trajectory modeling.                                                                                                                                                     |
| Modern Exterior Ballistics, The Launch<br>and Flight Dynamics of Symmetric<br>Projectiles.<br>Robert L. McCoy,<br>Shiffer Publishing Ltd., 1999<br>ISBN: 0-7643-0720-7                                                                                                                                                                                                                                           | Covers the free flight dynamics<br>of symmetric projectiles. Provides<br>many examples of projectile motion<br>illustrating key flight behaviors.                        |

ANNEX I to  
STANAG 4355  
Edition 3

International Standard ISO 2533:  
Standard Atmosphere, First Edition  
International Organization for  
Standardization, 1975.

Specifies the characteristics of an ISO  
Standard Atmosphere and is intended for use  
in calculations and design of flying vehicles.  
Also recommended in the processing of data  
from meteorological observations.